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- This product should be **used only when there is adult supervision present** as young children lack necessary judgment regarding safety and the consequences of product misuse.
- This product contains small parts and parts, which are sharp. This product contains electrically conductive parts. **Use caution with electrically conductive parts near or around power supplies, batteries and powered (live) circuits.**
- When the product is turned ON, activated or tested, some parts will move or rotate. **To avoid injuries to hands and fingers keep them away from any moving parts!**
- It is possible that an improperly connected or shorted circuit may cause overheating. **Should this happen, immediately disconnect the power supply or remove the batteries and do not touch anything until it cools down!** When everything is safe and cool, review the product tutorial to identify the cause.
- Only operate the product in accordance with the instructions and guidelines of this tutorial, otherwise parts may be damaged or you could be injured.
- Store the product in a cool dry place and avoid exposing the product to direct sunlight.
- After use, always turn the power OFF and remove or unplug the batteries before storing.

About Freenove

Freenove provides open source electronic products and services worldwide.

Freenove is committed to assist customers in their education of robotics, programming and electronic circuits so that they may transform their creative ideas into prototypes and new and innovative products. To this end, our services include but are not limited to:

- Educational and Entertaining Project Kits for Robots, Smart Cars and Drones
- Educational Kits to Learn Robotic Software Systems for Arduino, Raspberry Pi and micro: bit
- Electronic Component Assortments, Electronic Modules and Specialized Tools
- **Product Development and Customization Services**

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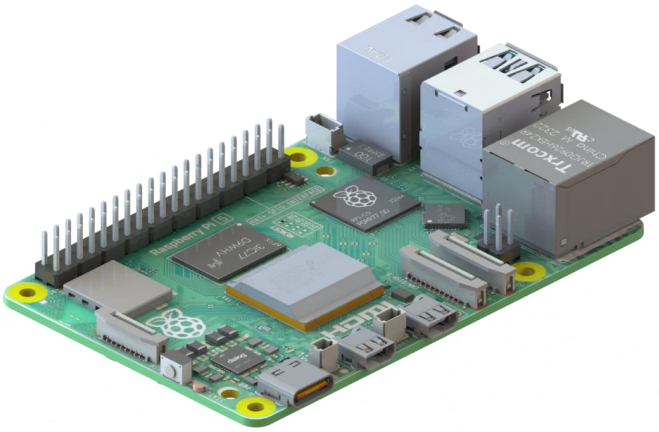
List

Raspberry Pi 5 x 1 
27W Power Adapter x 1 （or a power adapter compatible with Raspberry Pi official one that can output 5.1V/5A）
Micro SD Card (TF Card), Card Reader x 1
NVMe SSD x1

Installing Raspberry Pi OS

After the assembly is complete, we will start installing the system for the Raspberry Pi. Regardless of whether you want to install the system on the NVMe SSD or not, you need to install the system on an SD card or USB flash drive first.

Analysis

1. First of all, make sure you can enter the Raspberry Pi OS via SD card or U drive.
2. After booting the Raspberry Pi, you can use it to flash the OS image directly onto the NVMe SSD. Alternatively, you can purchase an NVMe SSD to USB adapter and flash the image using USB on Windows or macOS, much like you would for an SD card or USB drive.
3. With this analysis in mind, we can systematically carry out the necessary steps.

Raspberry Pi models manufactured at different times might not boot up in the same way as described, but that's okay; just follow our guide to proceed.
There are various ways to burn the Raspberry Pi OS to SSD, each requiring different hardware tools.

Ways	Ways of burning	Requirements
1	Use Raspberry Pi to burn the OS. This requires that you can successfully boot up the Raspberry Pi via SD card or U disk. (**Recommended, described in this tutorial**)	An SD card or a U disk that can access the Raspberry Pi OS.
2	Purchase an NVMe SSD to USB adapter and flash the image just like you would with an SD card or USB drive.	NVME SSD to USB adapter (need to be bought separately)
3	If there are spare M.2 NVME interface on the motherboard of your PC, you can insert the SSD to it to flash the OS.	PC with M.2 NVME interface

Caution: Incompatible SSDs

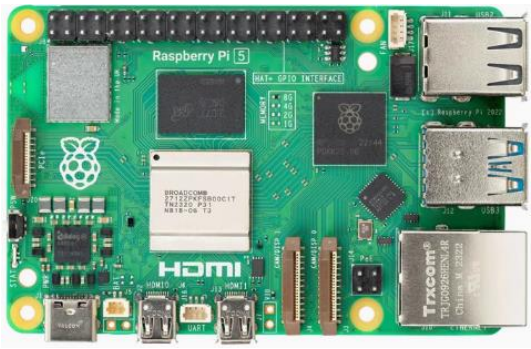
The recognition and read/write operations of the NVMe SSD are handled by the drivers of the Raspberry Pi 5. If you find that your SSD cannot be recognized by Pi 5 or is not readable/writable, please try to find a driver suitable for your SSD and install it on the Raspberry Pi, or replace the SSD, or purchase the adapter kit that comes with the SSD.

1 Flashing OS to SD Card or USB Drive

Based on the analysis above, our first step should be to install the Raspberry Pi operating system onto an SD card or USB drive, with a capacity of at least 16GB. If you are already able to boot the Raspberry Pi using an SD card or USB drive, you can skip this section and move on to the next section.

1.1 Component List

Required Components

Raspberry Pi 5 x 1 	One 27W power adapter (or a power adapter compatible with Raspberry Pi official one that can output 5.1V/5A)
Micro SD Card（TF Card）x1，Card Reader x1	

1.2 Raspberry Pi OS

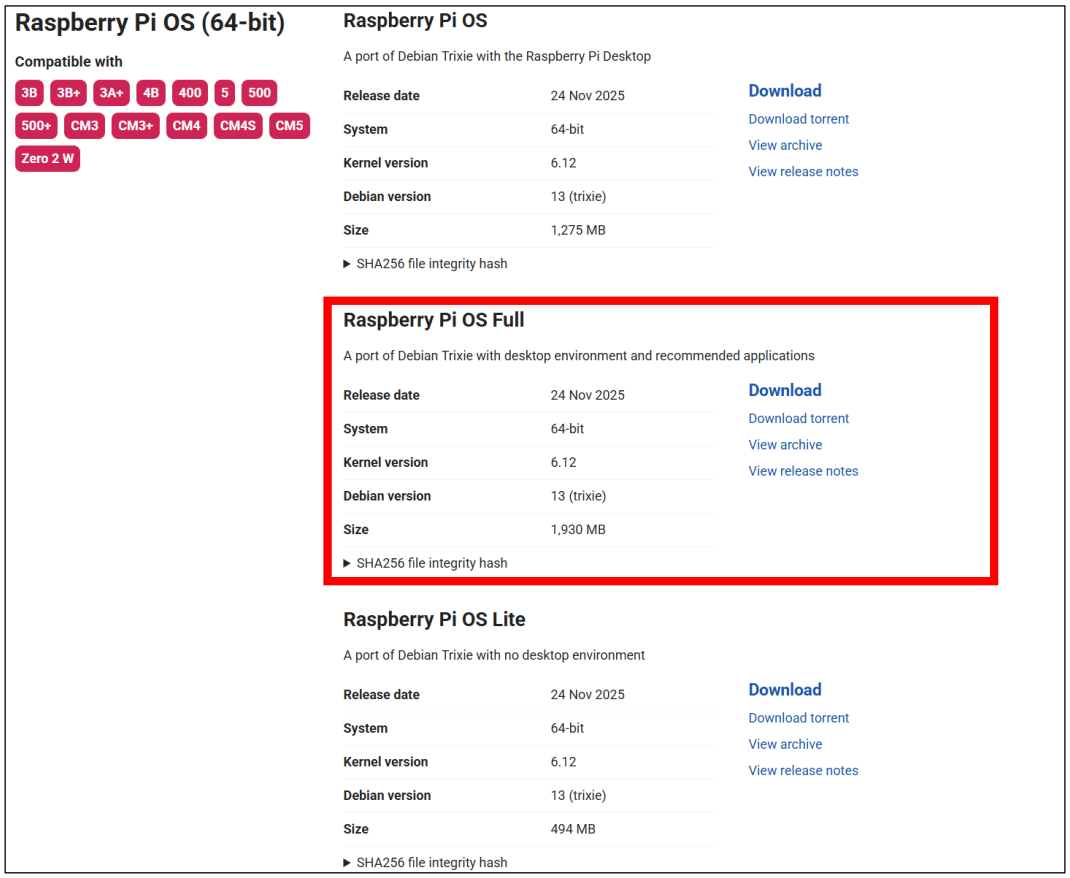
Without Screen - Use Raspberry Pi - under Windows PC: https://youtu.be/XpiT_ezb_7c
With Screen - Use Raspberry Pi - under Windows PC: https://youtu.be/HEywFsFrj3I

Automatically Method (Recommended)

You can follow the official method to install the system for raspberry pi via visiting link below:
https://projects.raspberrypi.org/en/projects/raspberry-pi-setting-up/2
In this way, the system will be downloaded **automatically** via the application.

Manually Methed

After installing the Imager Tool in the **link above**. You can **also** download the system **manually** first. (Note: Unless necessary, it is not recommended to manually download the offline installation package.)
Visit https://www.raspberrypi.com/software/operating-systems/

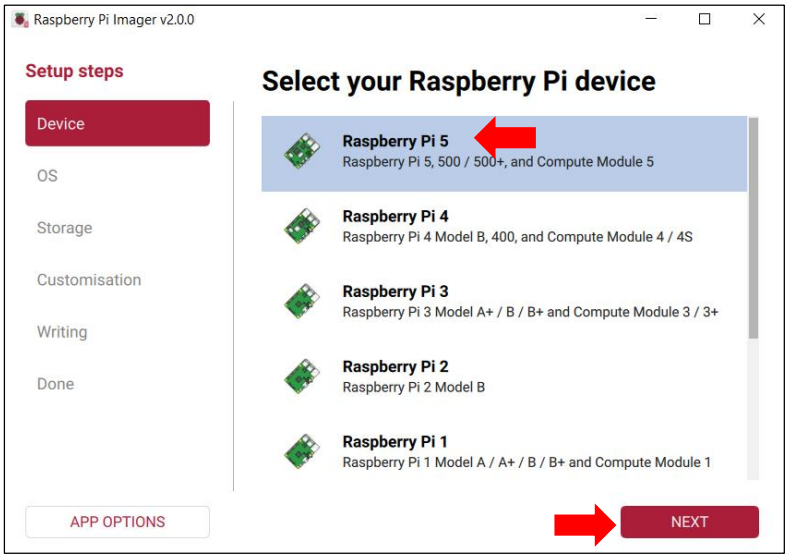


Choose Raspberry Pi Device

First, put your Micro **SD card** into card reader and connect it to USB port of PC.

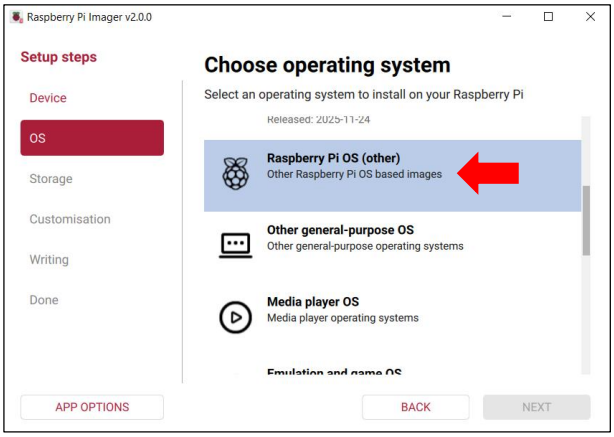
Open Raspberry Pi Imager.

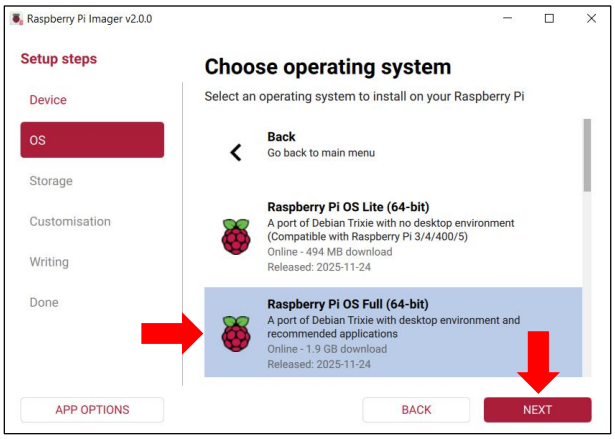
Select **Raspberry Pi 5** under Device, and then click NEXT.



Choose Operating System

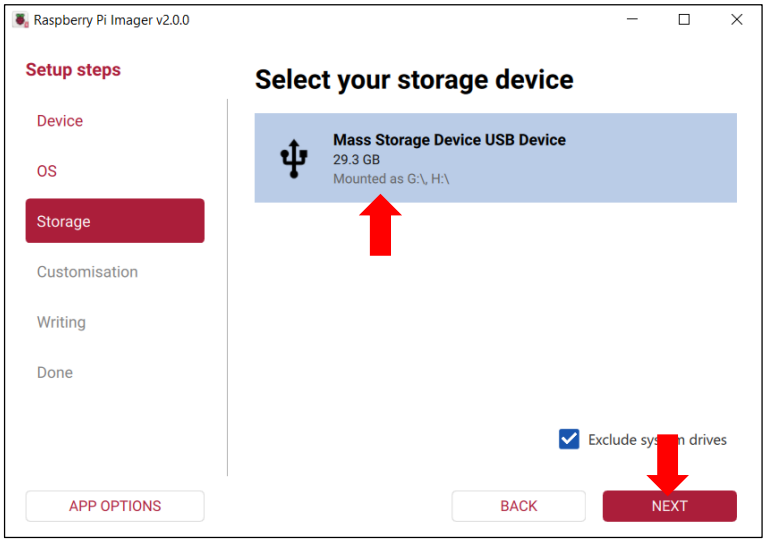
Click OS, scroll down to **locate and click Raspberry Pi OS (other)**, and then select **Raspberry Pi OS Full(64-bit)**. Click NEXT.





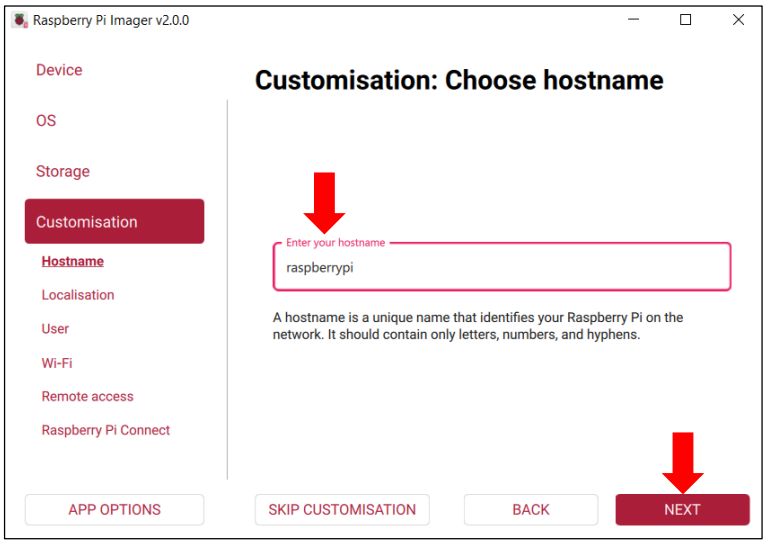
Choose Storage Devide

Choose the SD card and click on Next.

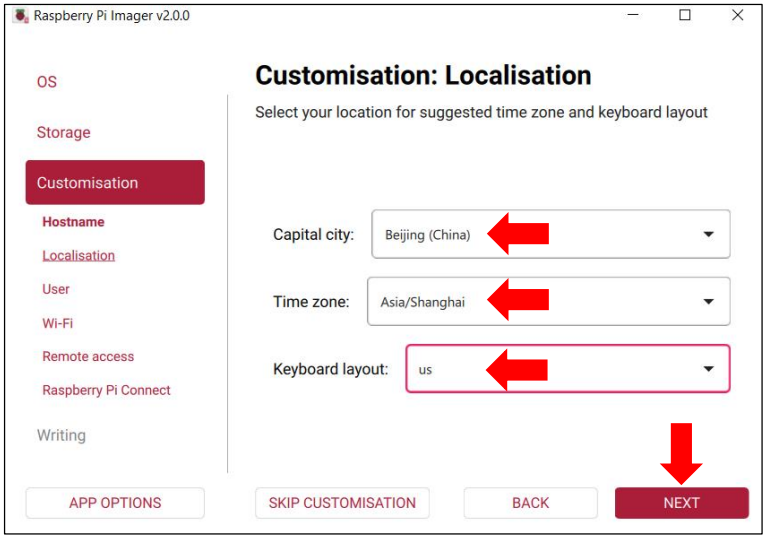


Customisation

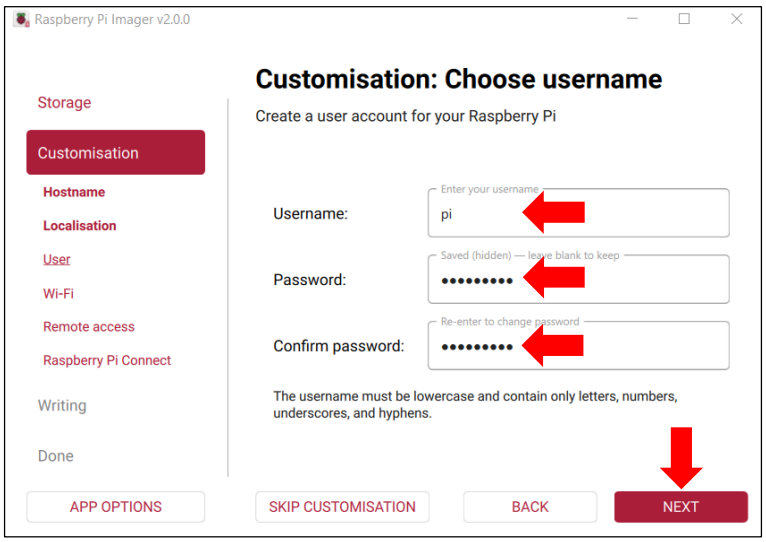
Step 1: Input the hostname, which is "**raspberrypi**" by default. Then click NEXT.



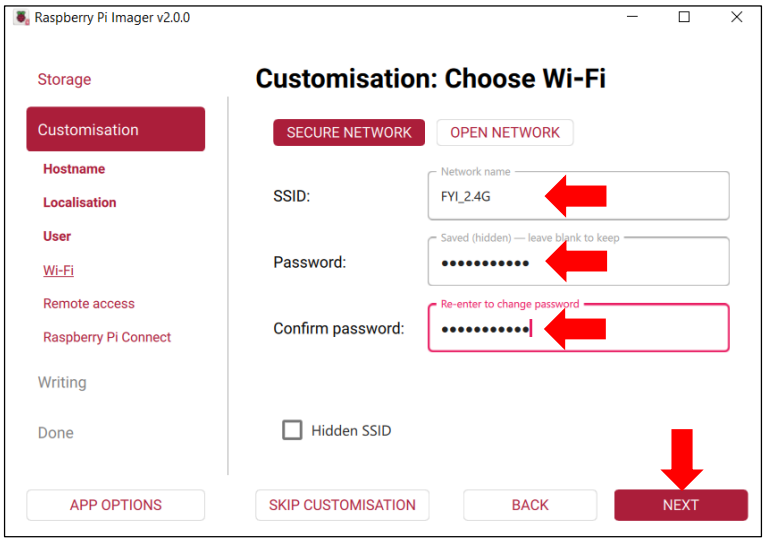
Step 2: Select your location, timezone and keyboard layout. Then click NEXT.



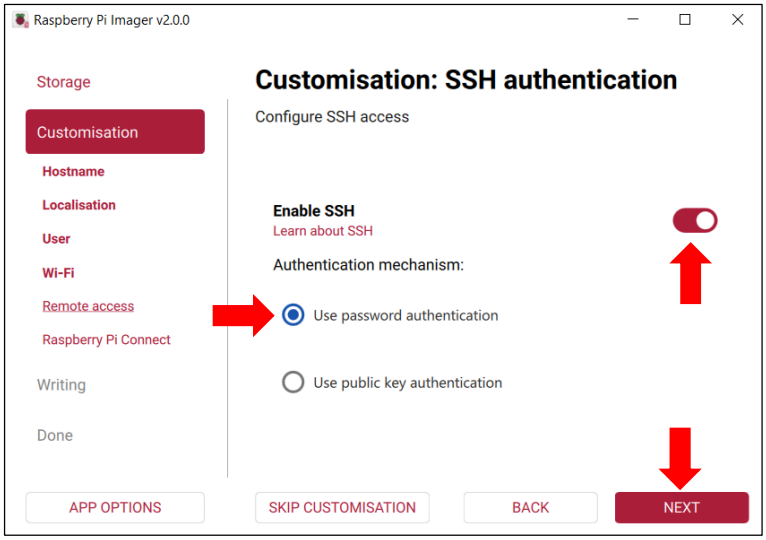
Step 3: Configure your username and passowrd. The password needs to re-enter to confirm. The default username is **pi** and the default password is **raspberry**. Then click NEXT.



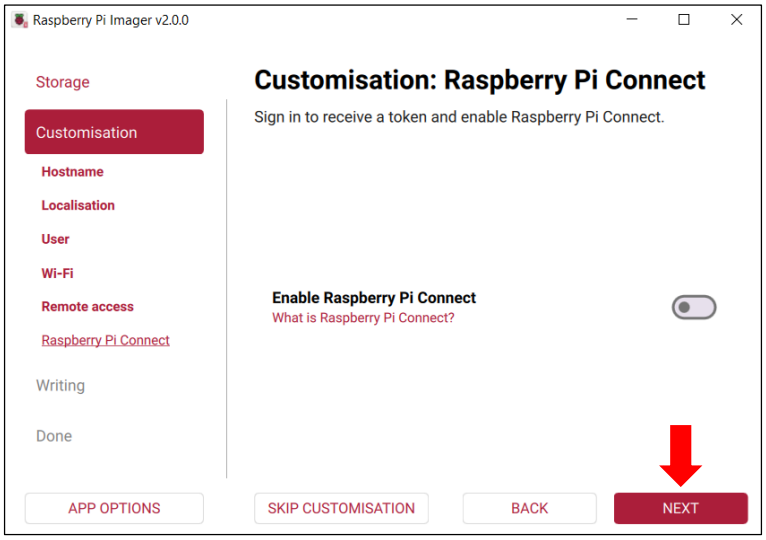
Step 4: Input the SSID and password of your wireless network. The passwords needs to re-enter to confirm. Then click NEXT.



Step 5: Enable SSH for ssh remote and click NEXT.

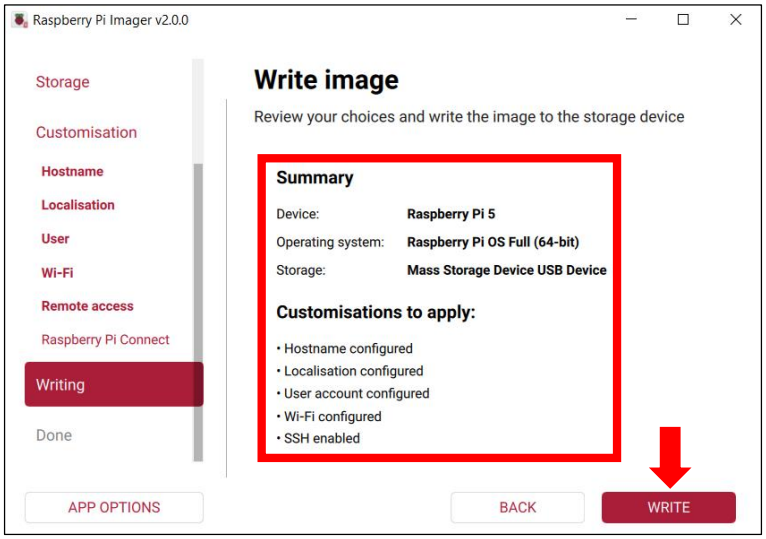


Step 6: Click NEXT.

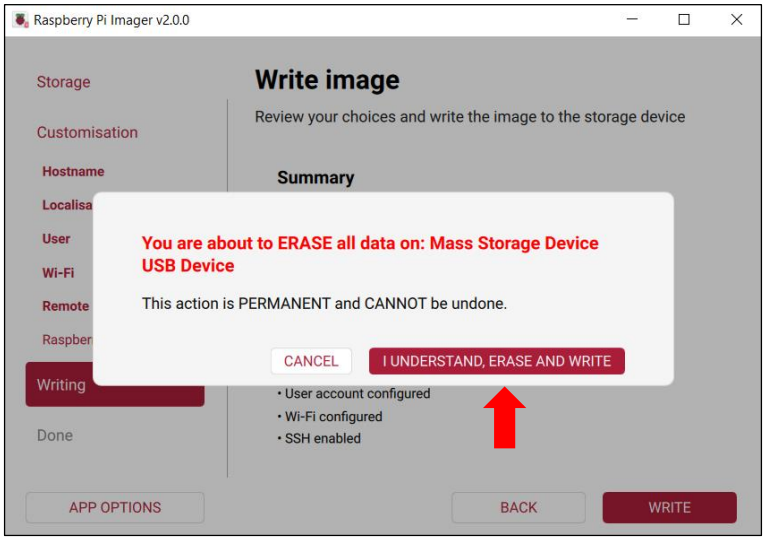


Writing System to Micro SD Card

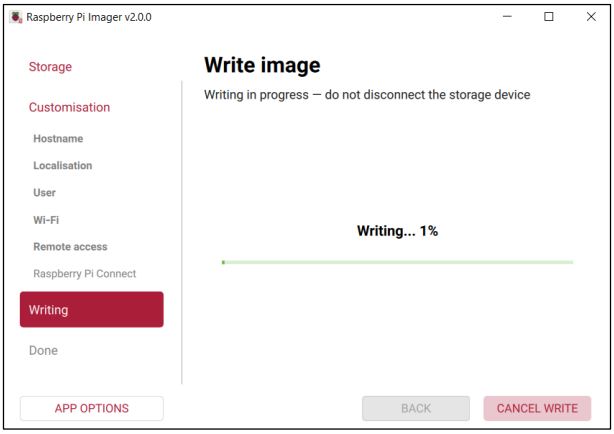
After finishing customisation configuration, the interface will show all the configuration you have made. Click WRITE to write the OS to your SD card after confirming everything is correct.

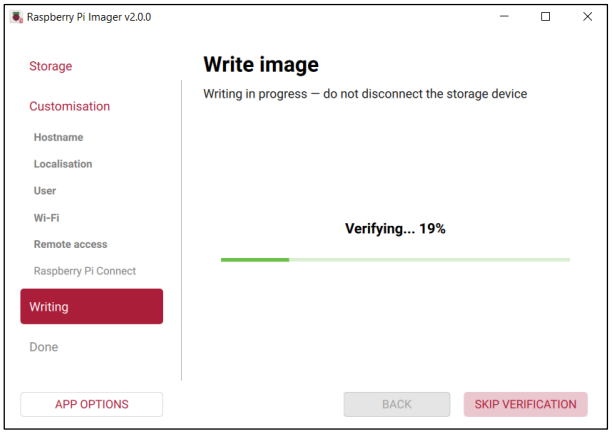


Warning: Before starting the write process, ensure you have backed up all data on the storage device. This operation will permanently erase all data on the device. If you have completed the backup, click **I UNDERSTAND, ERASE AND WRITE** to continue.



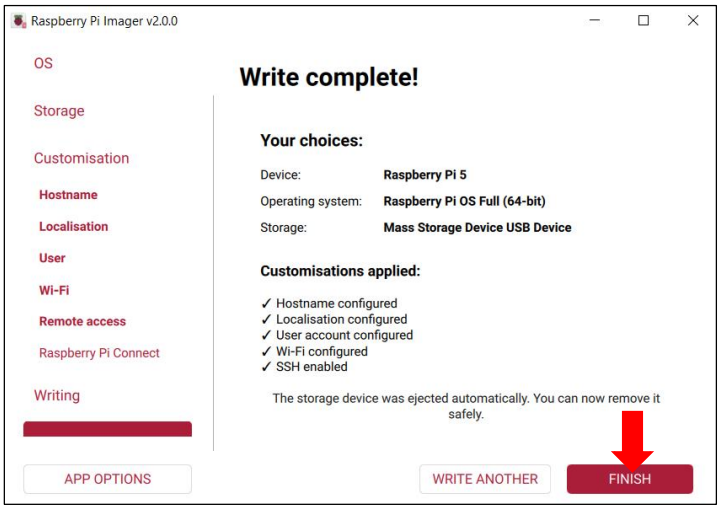
Wait for it to finish writing and verifying.





Done

Congratulations! The Raspberry Pi OS has been successfully written to the SD card. All your custom configurations have taken effect. Click FINISH to complete the process.



1.3 HDMI Monitor desktop

If you do not have a spare monitor, please skip to next section Remote desktop & VNC. If you have a spare monitor, please follow the steps in this section.

Connect the Raspberry Pi 5 to your display via the HDMI interface, attach your mouse and keyboard through the USB ports, and finally, connect your power supply (making sure that it meets the specifications required by your RPi Module Version. Your RPi should start (power up). Later, after setup, you will need to enter your user name and password to login. The default user name: pi; password: raspberry. After login, you should see the following screen.

Congratulations! You have successfully installed the RASPBERRY PI OS on your RPi.

Raspberry Pi 5 integrates a Wi-Fi adaptor. You can use it to connect to your Wi-Fi. Then you can use the wireless remote desktop to control your RPi. This will be helpful for the following work.

1.4 Remote desktop & VNC

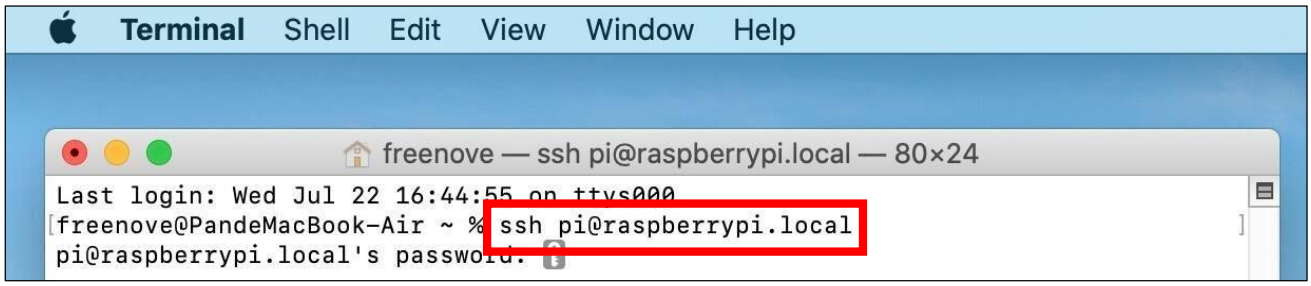
If you don't have a spare display, mouse and keyboard for your RPi, you can use a remote desktop to share a display, keyboard, and mouse with your PC. Below is how to use:
MAC OS remote desktop and Windows OS remote desktop.

Mac OS Remote desktop

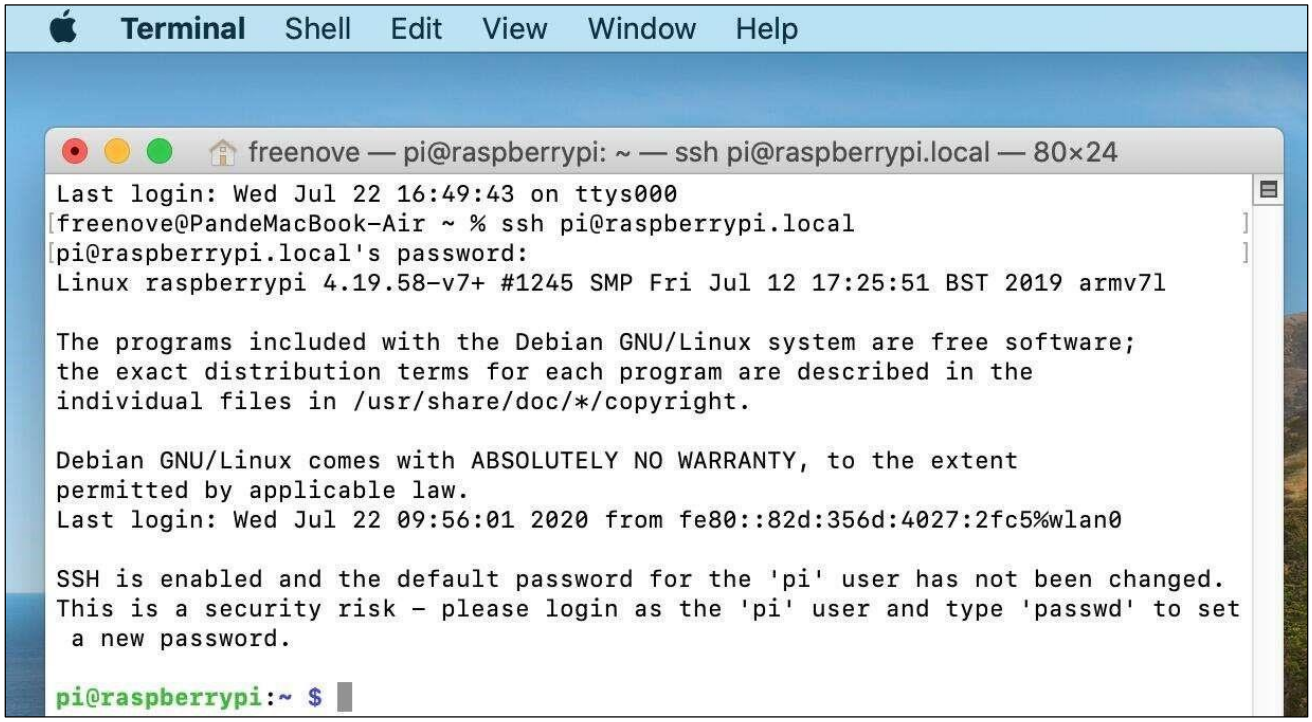
Open the terminal and type following command.
If you have set your own hostname during writing the OS, please modify it accordingly.
If this command doesn't work, please move to next page.

```
ssh pi@raspberrypi.local
```

The password is **raspberry** by default, case sensitive.



You may need to type **yes** during the process.

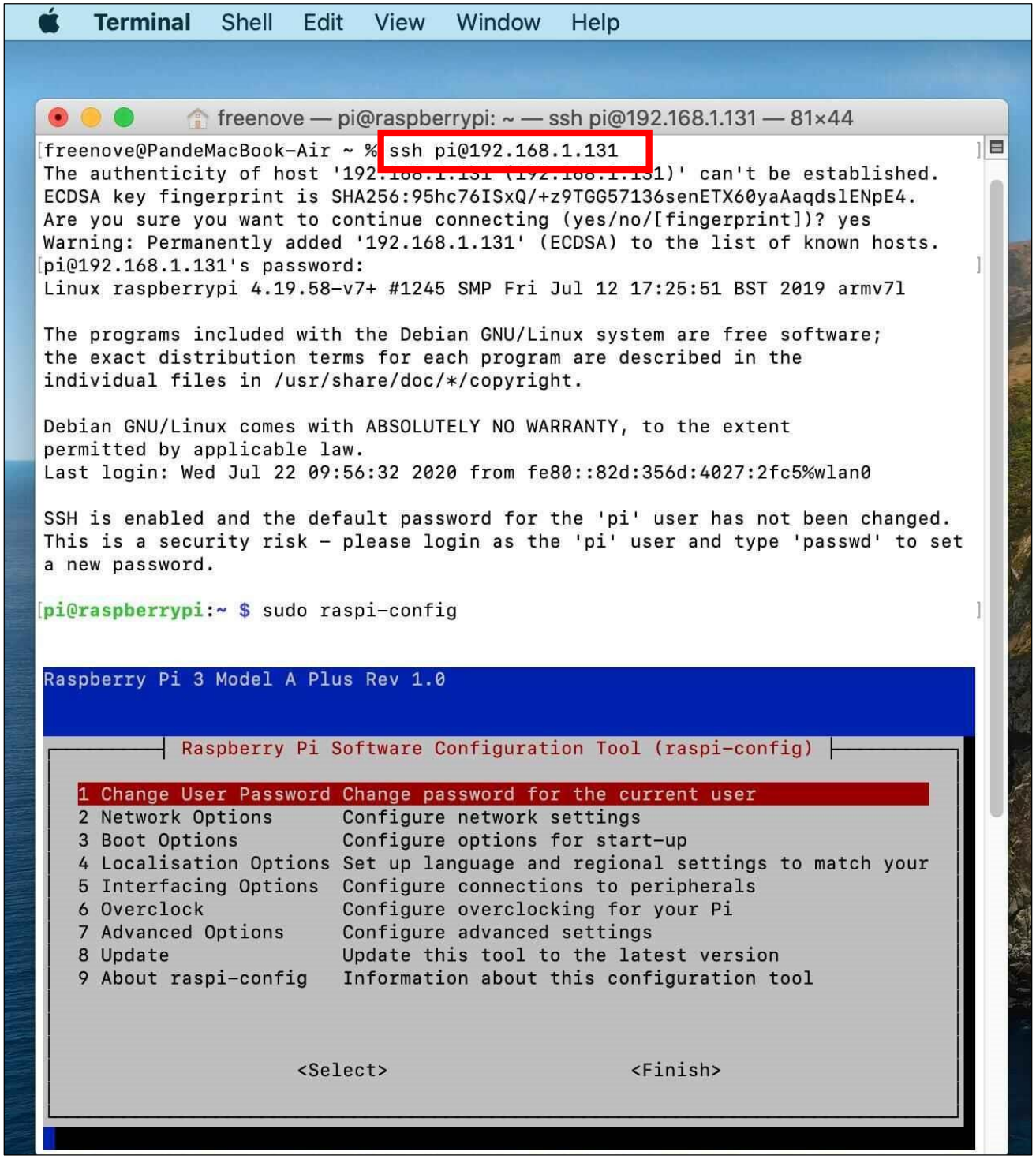


You can also use the IP address to log in Pi.
Enter **router** client to **inquiry IP address** named "raspberry pi". For example, I have inquired to **my RPi IP address, and it is "192.168.1.131".**

Open the terminal and type following command.

```
ssh pi@192.168.1.131
```

When you see **pi@raspberrypi:~ $**, you have logged in Pi successfully. Then you can skip to next section.



Then you can skip to VNC Viewer.

Windows OS remote desktop

If you are using win10, you can use follow way to login Raspberry Pi without desktop.

Press **Win+R**. Enter **cmd**. Then use this command to check IP:

```
ping -4 raspberrypi.local
```

```
Command Prompt
Microsoft Windows [Version 10.0.19044.2130]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ping -4 raspberrypi.local

Pinging raspberrypi.local [192.168.1.147] with 32 bytes of data:
Reply from 192.168.1.147: bytes=32 time=10ms TTL=64
Reply from 192.168.1.147: bytes=32 time=4ms TTL=64
Reply from 192.168.1.147: bytes=32 time=124ms TTL=64
Reply from 192.168.1.147: bytes=32 time=7ms TTL=64

Ping statistics for 192.168.1.147:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 124ms, Average = 36ms
```

Then 192.168.1.147 is my Raspberry Pi IP.

Or enter **router** client to **inquiry IP address** named "**raspberrypi**". For example, I have inquired to **my RPi IP address, and it is "192.168.1.147".**

```
ssh pi@xxxxxxxxxxx(IP address)
```

Enter the following command:

```
ssh pi@192.168.1.147
```

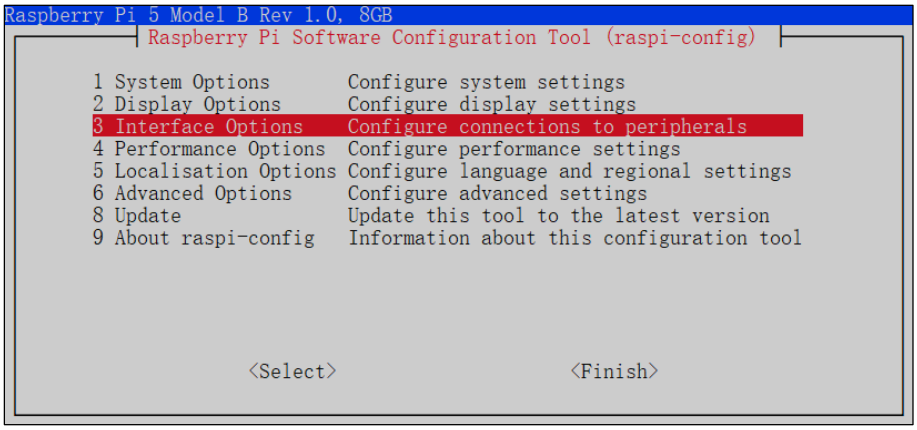
```
C:\Users\Administrator>ssh pi@192.168.1.147
pi@192.168.1.147's password:
Linux raspberrypi 5.15.74-v7+ #1595 SMP Wed Oct 26 11:03:05 BST 2022 armv7l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Nov  7 10:19:19 2022 from 192.168.1.127

SSH is enabled and the default password for the 'pi' user has not been changed.
This is a security risk - please login as the 'pi' user and type 'passwd' to set a new password.

pi@raspberrypi:~ $
```

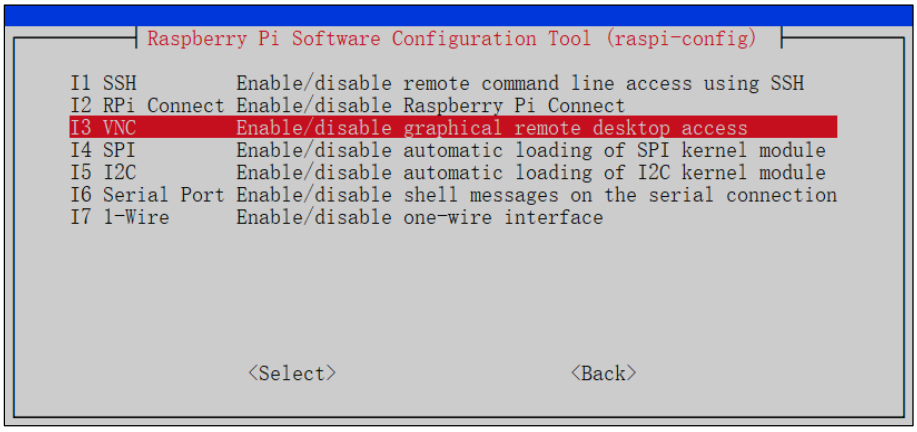
1.5 VNC Viewer & VNC

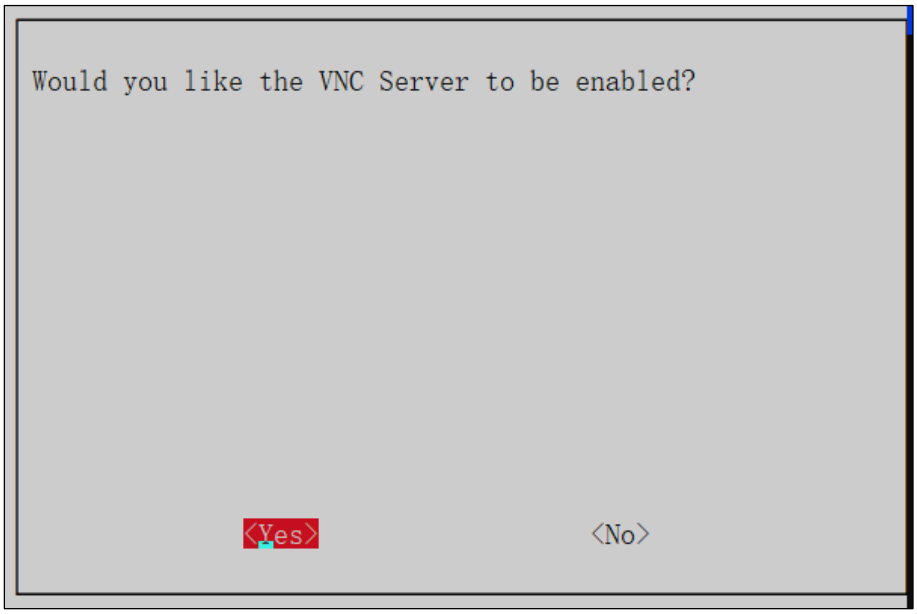
Enable VNC

Enter the following commands: select Interface Options -> I3 VNC -> Enter -> Yes -> OK. You may need to restart the Raspberry Pi 5 here, select OK, and then open the VNC interface.

```
sudo raspi-config
```

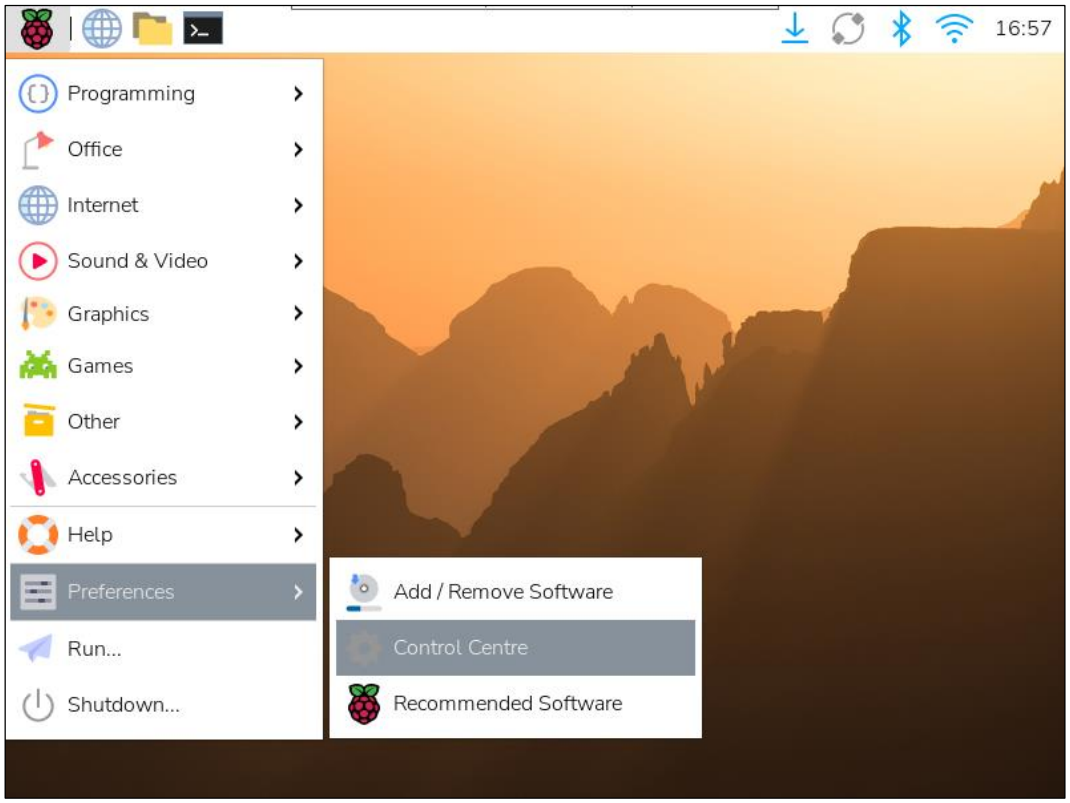


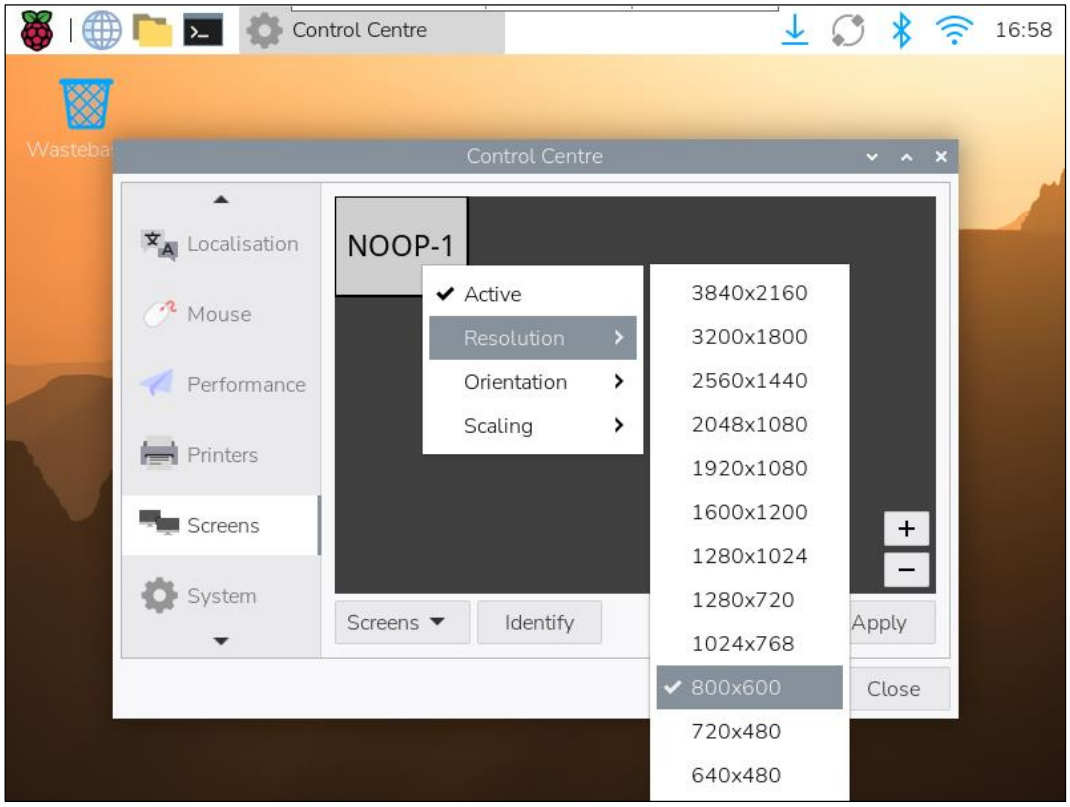




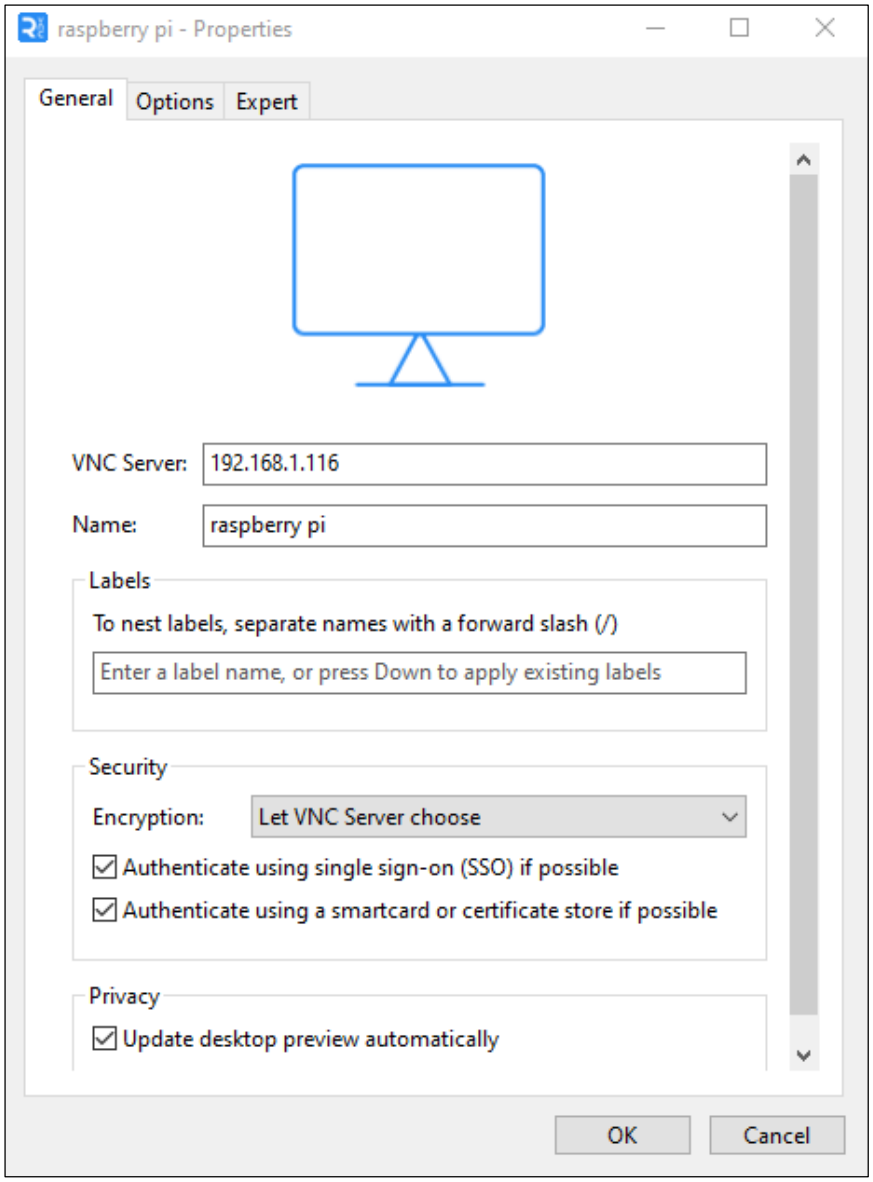
Set Resolution

You can also set other resolutions. If you don't know what to set, you can set it as 800x600 first.



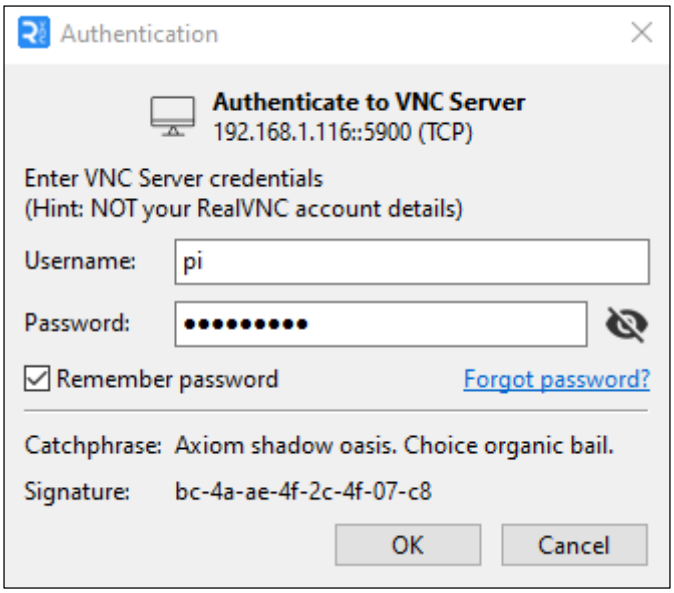


Then download and install VNC Viewer according to your computer system by click following link:
https://www.realvnc.com/en/connect/download/viewer/
After installation is completed, open VNC Viewer. And click File -> New Connection. Then the interface is shown below.

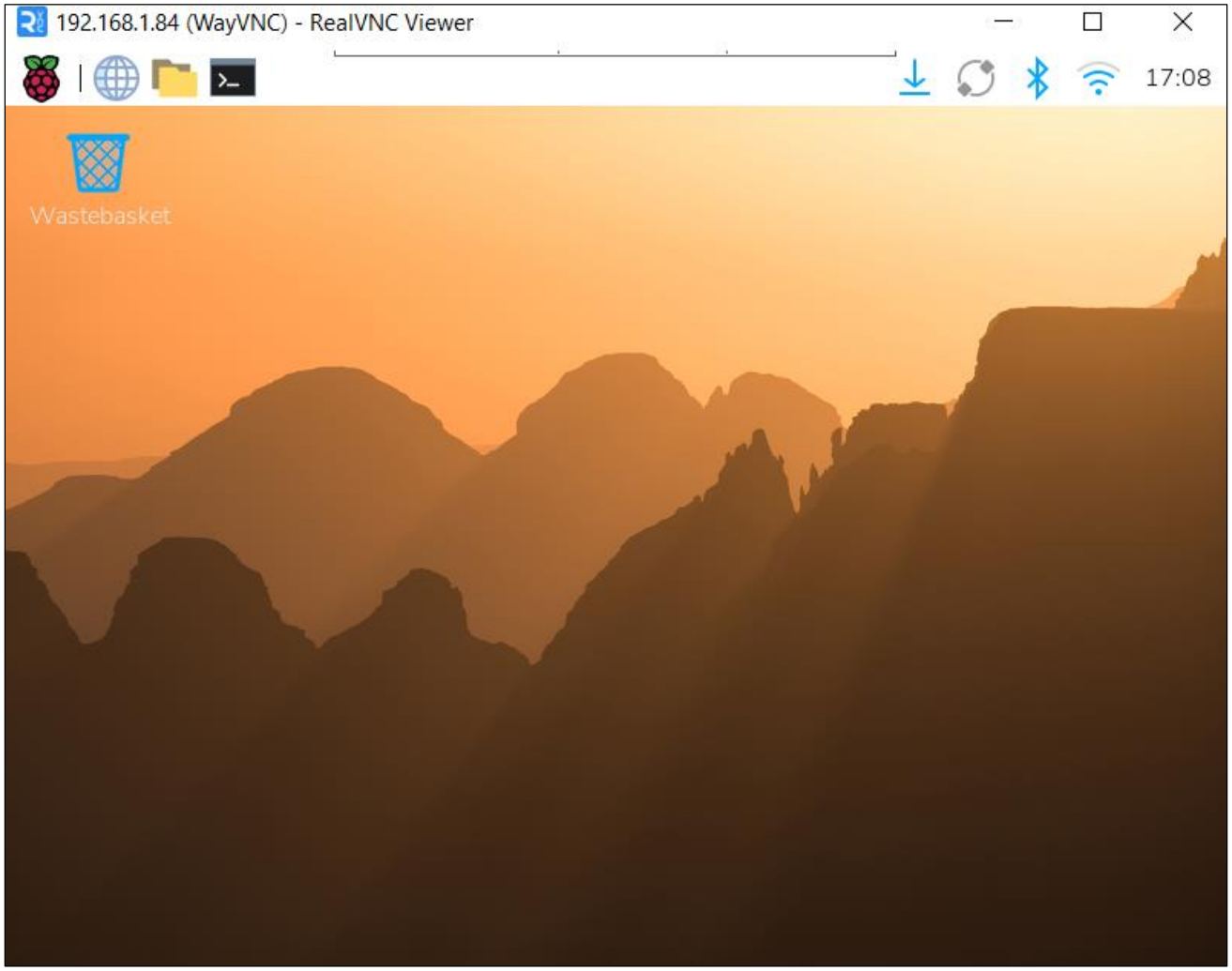


Enter ip address of your Raspberry Pi and fill in a name. Then click OK.
Then on the VNC Viewer panel, double-click new connection you just created,

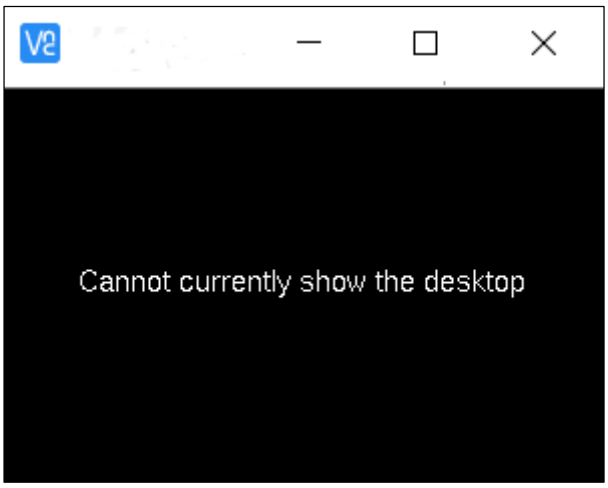
Then the following dialog box pops up.
Enter username: **pi** and Password: **raspberry**.
If you have set your own username and password, please input them accordingly. Then click OK.



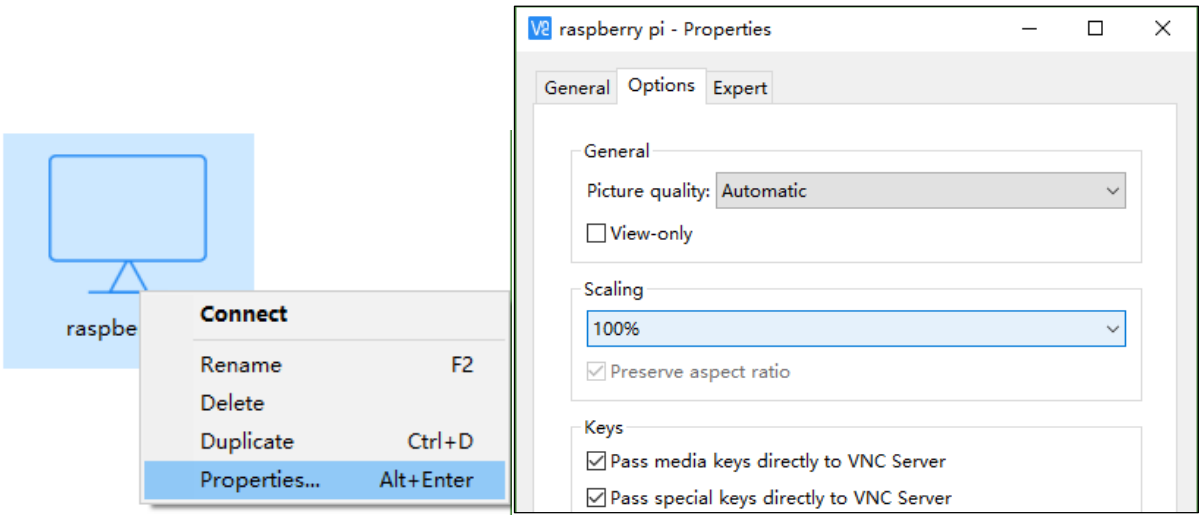
Here, you have logged in to Raspberry Pi successfully by using VNC Viewer.



If there is black window, please **set another resolution**.



In addition, your VNC Viewer window may zoom your Raspberry Pi desktop. You can change it. On your VNC View control panel, click right key. And select Properties->Options label->Scaling. Then set proper scaling.



Here, you have logged in to Raspberry Pi successfully by using VNC Viewer and operated proper setting.

Raspberry Pi 5 integrates a Wi-Fi adaptor.If you did not connect Pi to WiFi. You can connect it to wirelessly control the robot.

2 Flashing OS to NVMe SSD

This step is to install the Raspberry Pi OS into the NVME SSD.

2.1 SSD Detection

(Note: Not all SSDs are supported by Pi5.)

Booting the Rpi 5 from SD card, and run the following command in the Terminal to check whether SSD is detected.

Note that the information varies among SSDs.

lspci

```
pi@raspberrypi:~ $ lspci
0000:00:00.0 PCI bridge: Broadcom Inc. and subsidiaries Device 2712 (rev 21)
0000:01:00.0 Non-Volatile memory controller: Silicon Motion, Inc. SM2263EN/SM2263XT SSD Controller (rev 03)
0001:00:00.0 PCI bridge: Broadcom Inc. and subsidiaries Device 2712 (rev 21)
0001:01:00.0 Ethernet controller: Device 1de4:0001
```

lsblk

```
pi@raspberrypi:~ $ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
mmcblk0     179:0    0  29.8G  0 disk
├mmcblk0p1 179:1    0   512M  0 part /boot/firmware
└mmcblk0p2 179:2    0  29.3G  0 part /
nvme0n1     259:0    0 476.9G  0 disk
```

As shown in the figure above, the device 'nvme0n1' with a capacity of 476.9GBytes shows up, indicating that the SSD has been correctly recognized. The detected capacity will depend on the size of your SSD. If your drive has been previously partitioned, you may also see some partition information displayed.

Please note: Installing the system will format the SSD, erasing all data. If necessary, please back up any data on your SSD before proceeding.

2.2 Enable PCIe 3.0(on OS written into SD Card)

If your SSD with Phison controller, you may need to enable PCIE 3.0. (This step is strongly recommended; without this step, the later process may fail.)

If it is not with Phison controller, you do not need to enable PCIE 3.0. You may skip this section.

Run the command lspci to check the controller.

```
pi@raspberrypi:~ $ lspci
0000:00:00.0 PCI bridge: Broadcom Inc. and subsidiaries BCM2712 PCIe Bridge (rev 21)
0000:01:00.0 Non-Volatile memory controller: Phison Electronics Corporation PS5021-E21
 PCIe4 NVMe Controller (DRAM-less) (rev 01)
0001:00:00.0 PCI bridge: Broadcom Inc. and subsidiaries BCM2712 PCIe Bridge (rev 21)
0001:01:00.0 Ethernet controller: Raspberry Pi Ltd RP1 PCIe 2.0 South Bridge
```

(The above screenshot is the result of a 128GB SSD with Phison as main controller.)

Enable PCIe Gen 3.0

Add the line dtparam=pciex1_gen=3 to /boot/firmware/config.txt to enable PCIe Gen3.0.

As shown below, enter the command to open the file.

```
sudo nano /boot/firmware/config.txt
```

Add the line dtparam=pciex1_gen=3 to the end of the file, as shown below:

Press Ctrl-O to save the file, Enter to confirm, and Ctrl-X to exit.

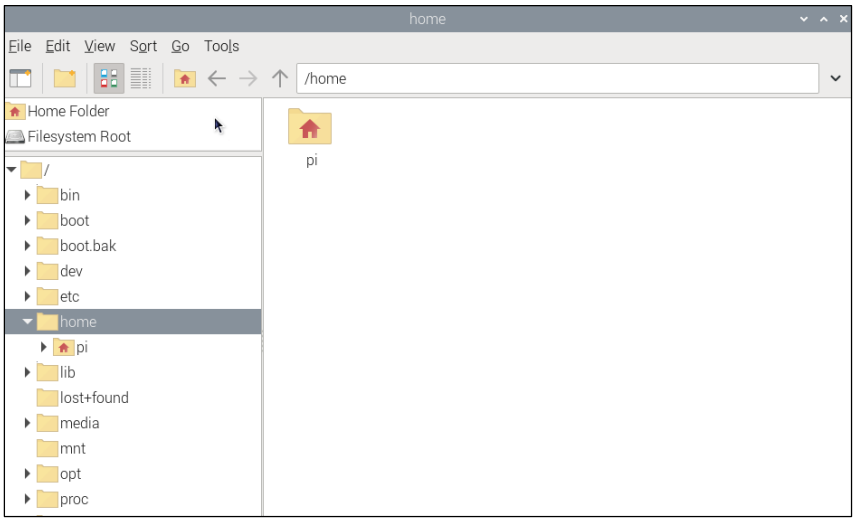
Reboot your Raspberry Pi.

```
sudo reboot
```

2.3 SSD Partitioning and Formatting

This step is not a must-do, but it can further test whether the SSD perform normally on Raspberry Pi to ensure smooth performance in later steps.

At this point, the hard drive cannot be seen in the file manager as the disk has not been partitioned yet.



Install a disk management tool with the following command:

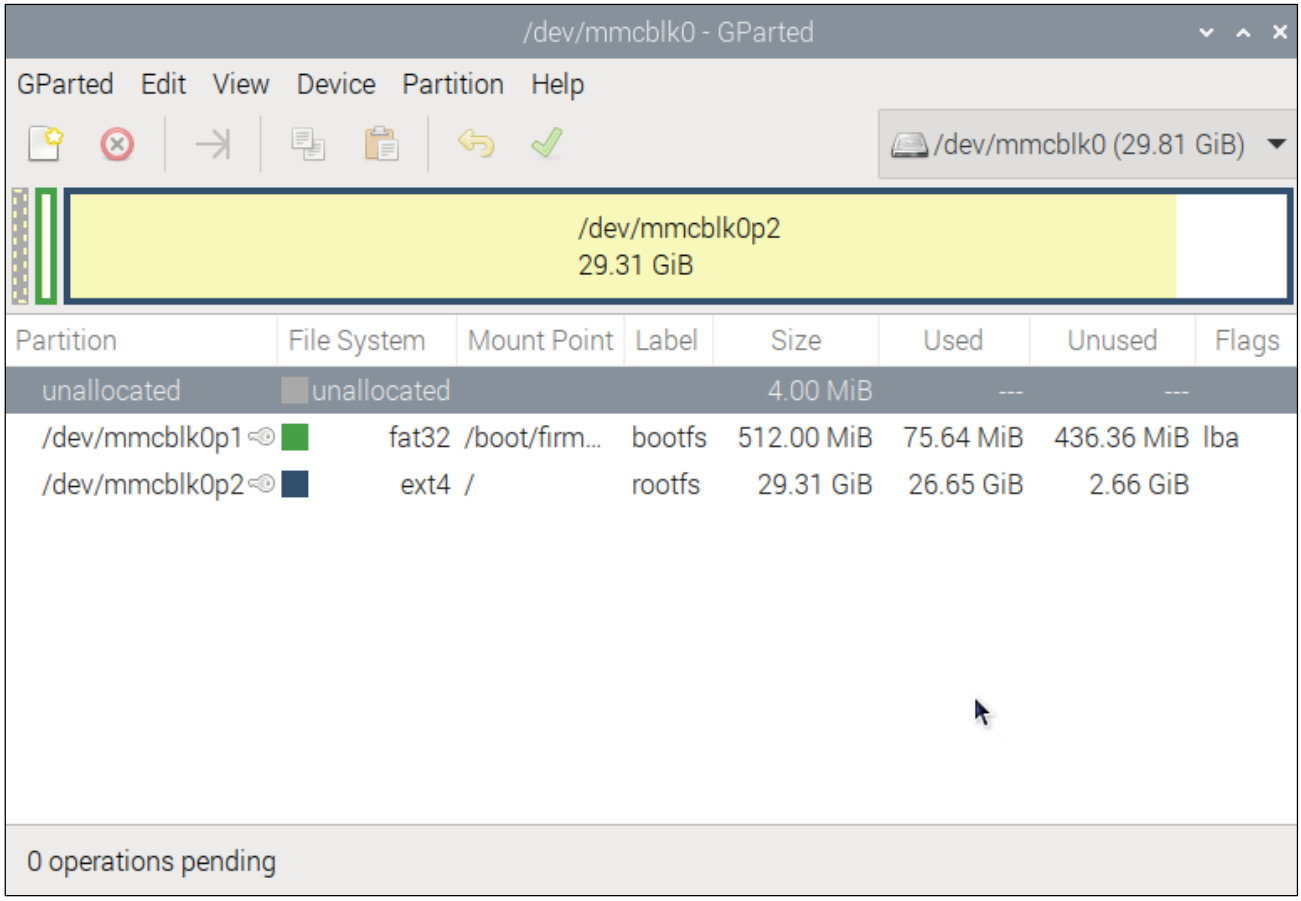
```
sudo apt-get install gparted
```

```
The following NEW packages will be installed:
  gparted gparted-common
0 upgraded, 2 newly installed, 0 to remove and 164 not upgraded.
Need to get 772 kB/2,483 kB of archives.
After this operation, 8,638 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://deb.debian.org/debian bookworm/main arm64 gparted arm64 1.3.1-1 [77
2 kB]
Fetched 772 kB in 0s (2,579 kB/s)
Selecting previously unselected package gparted-common.
(Reading database ... 222757 files and directories currently installed.)
Preparing to unpack .../gparted-common_1.3.1-1_all.deb ...
Unpacking gparted-common (1.3.1-1) ...
Selecting previously unselected package gparted.
Preparing to unpack .../gparted_1.3.1-1_arm64.deb ...
Unpacking gparted (1.3.1-1) ...
Setting up gparted-common (1.3.1-1) ...
Setting up gparted (1.3.1-1) ...
Processing triggers for mailcap (3.70+nmu1) ...
Processing triggers for desktop-file-utils (0.26-1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1.1) ...
Processing triggers for man-db (2.11.2-2) ...
pi@raspberrypi:~ $
```

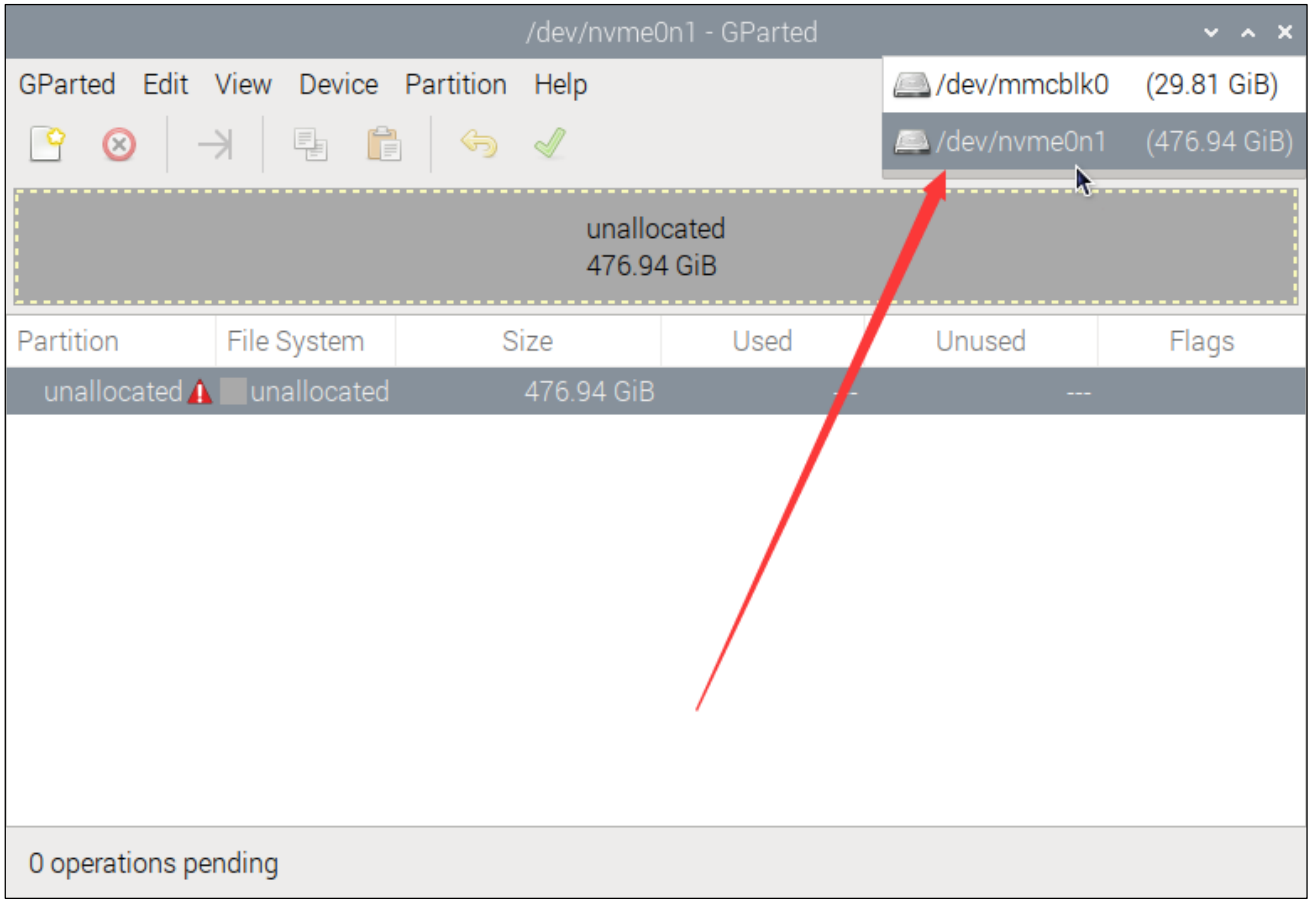
Open gparted with the command:

```
sudo gparted
```

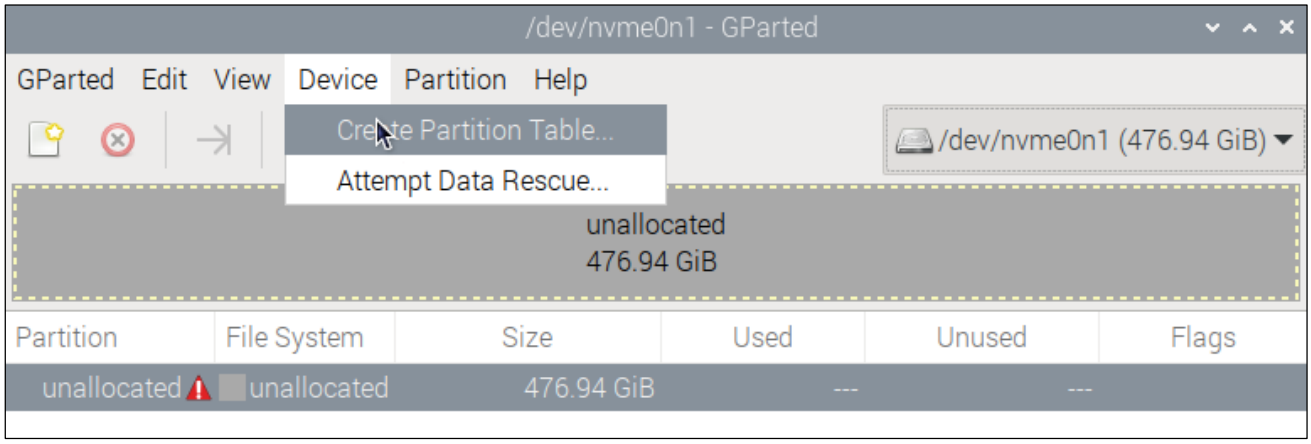
```
pi@raspberrypi:~ $ sudo gparted
error: XDG_RUNTIME_DIR is invalid or not set in the environment.
GParted 1.3.1
configuration --enable-libparted-dmraid --enable-online-resize
libparted 3.5
/dev/nvme0n1: unrecognised disk label
```



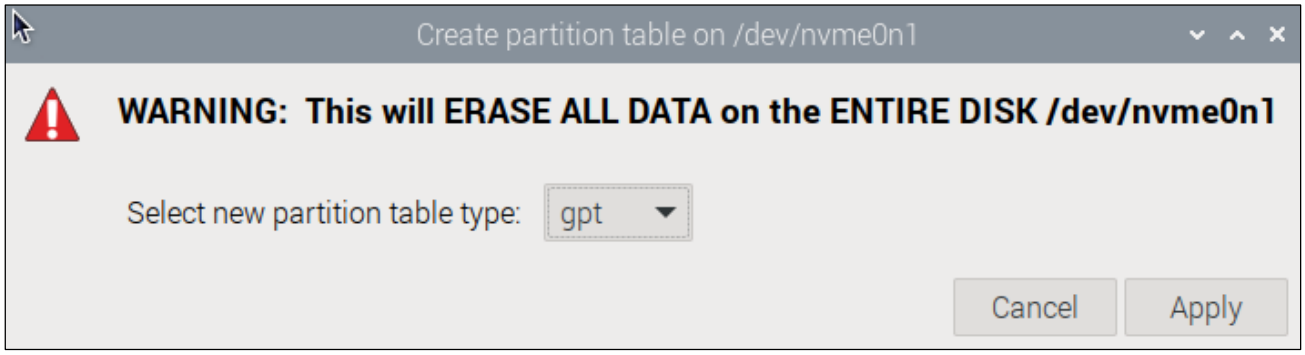
Click on the dropdown menu in the upper right corner and switch to NVME SSD.



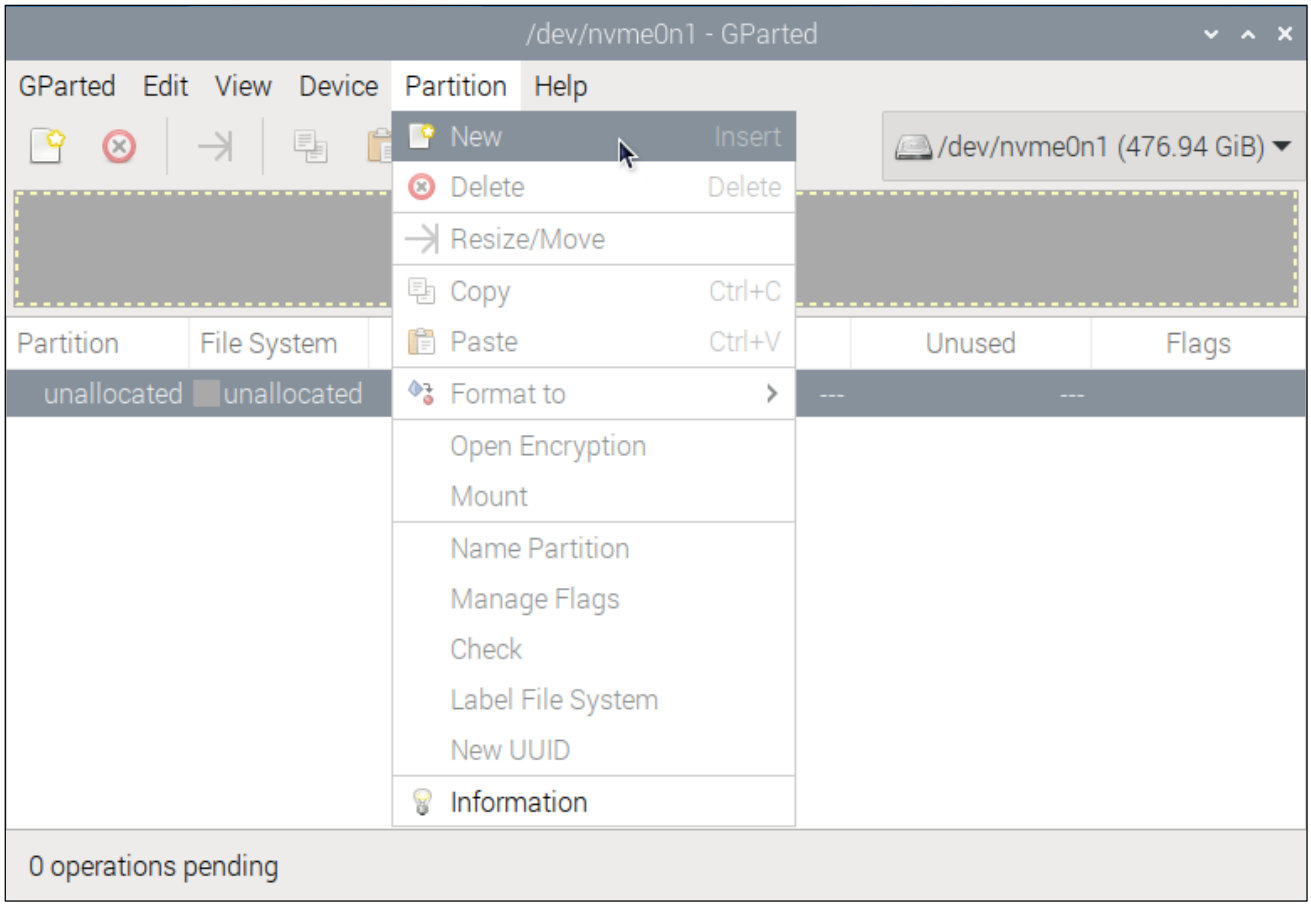
Click Device on the menu bar and select Create Partition Table.



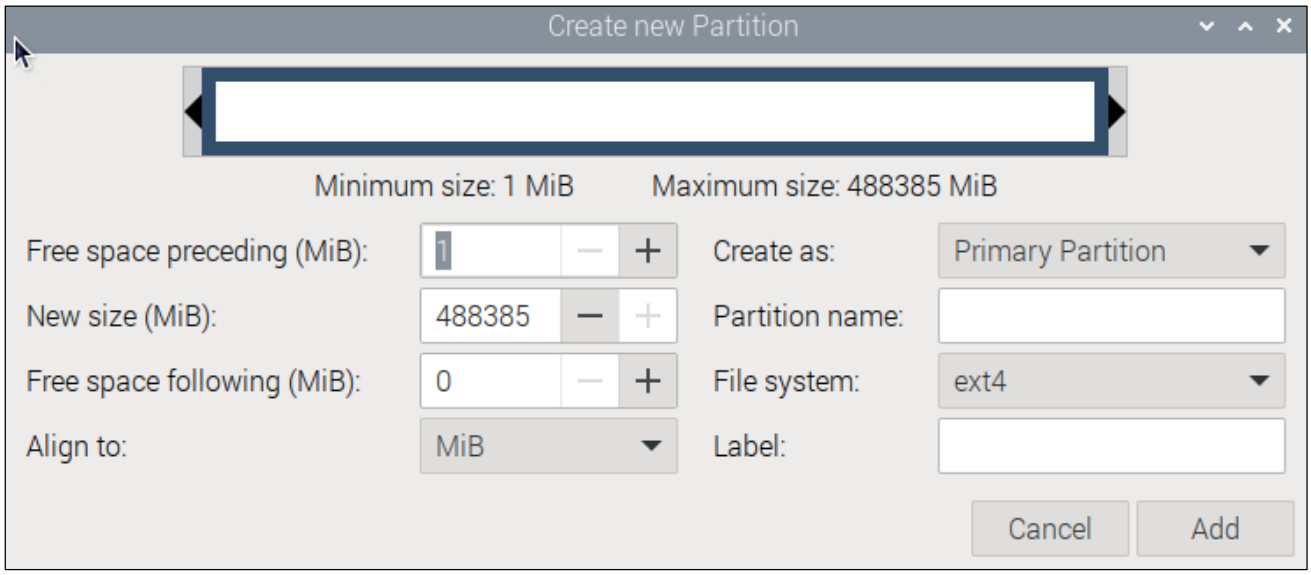
You will see the prompt that data will be erased. It is recommended to select gpt for partition table type. Click Apply.



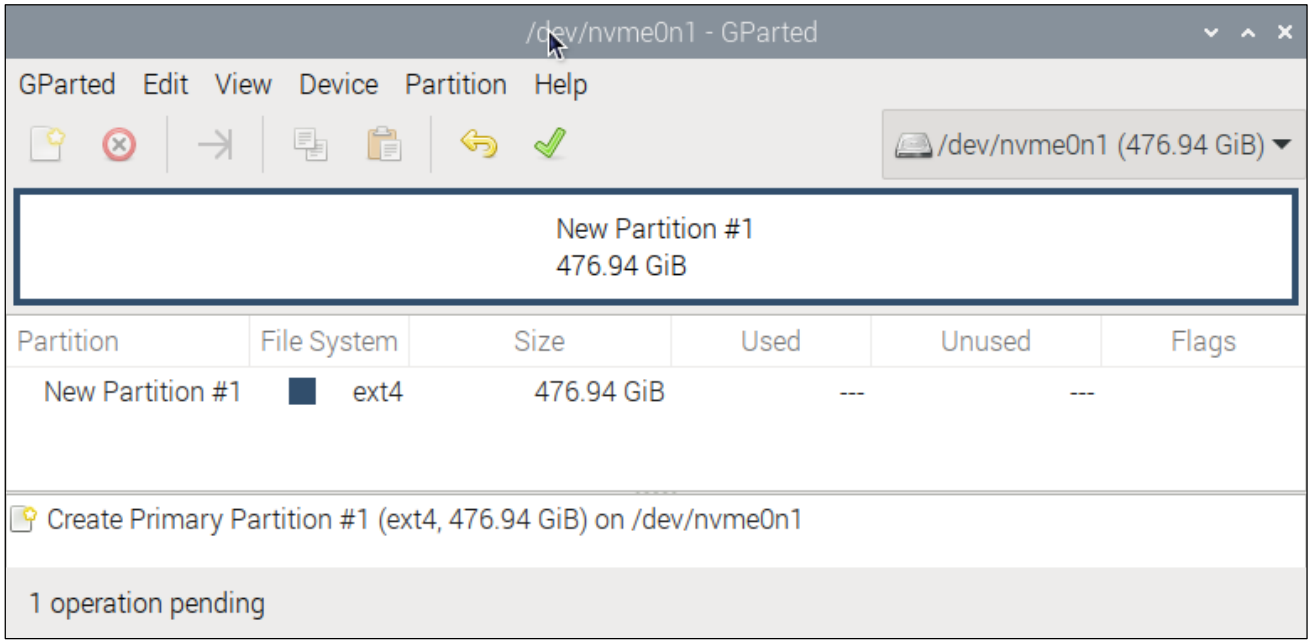
Click Partition on the menu bar, choose New.



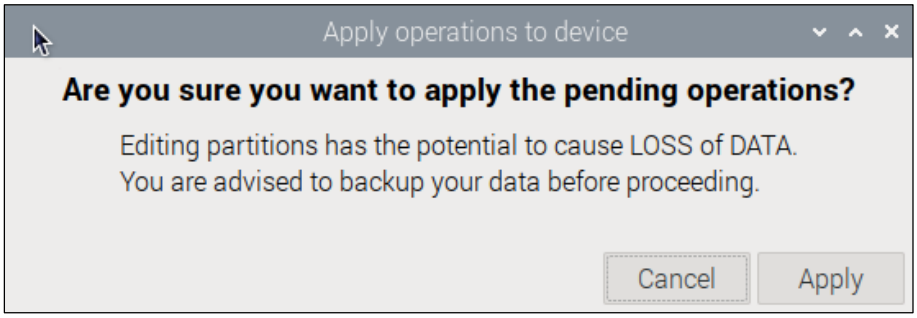
As shown in the figure below, the size of partition can be adjusted by dragging the mouse left and right, or by entering the size directly. The other options can be left as default setting. Here, we allocate all the capacity to a single partition. Click on Add.



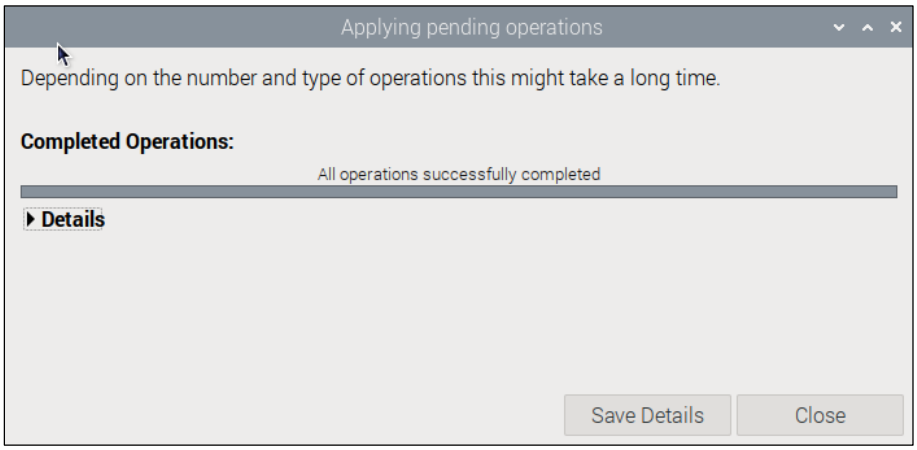
Click the check icon to save the partition just built, as illustrated below.



Click on Apply.



Wait for it to complete and click on Close.

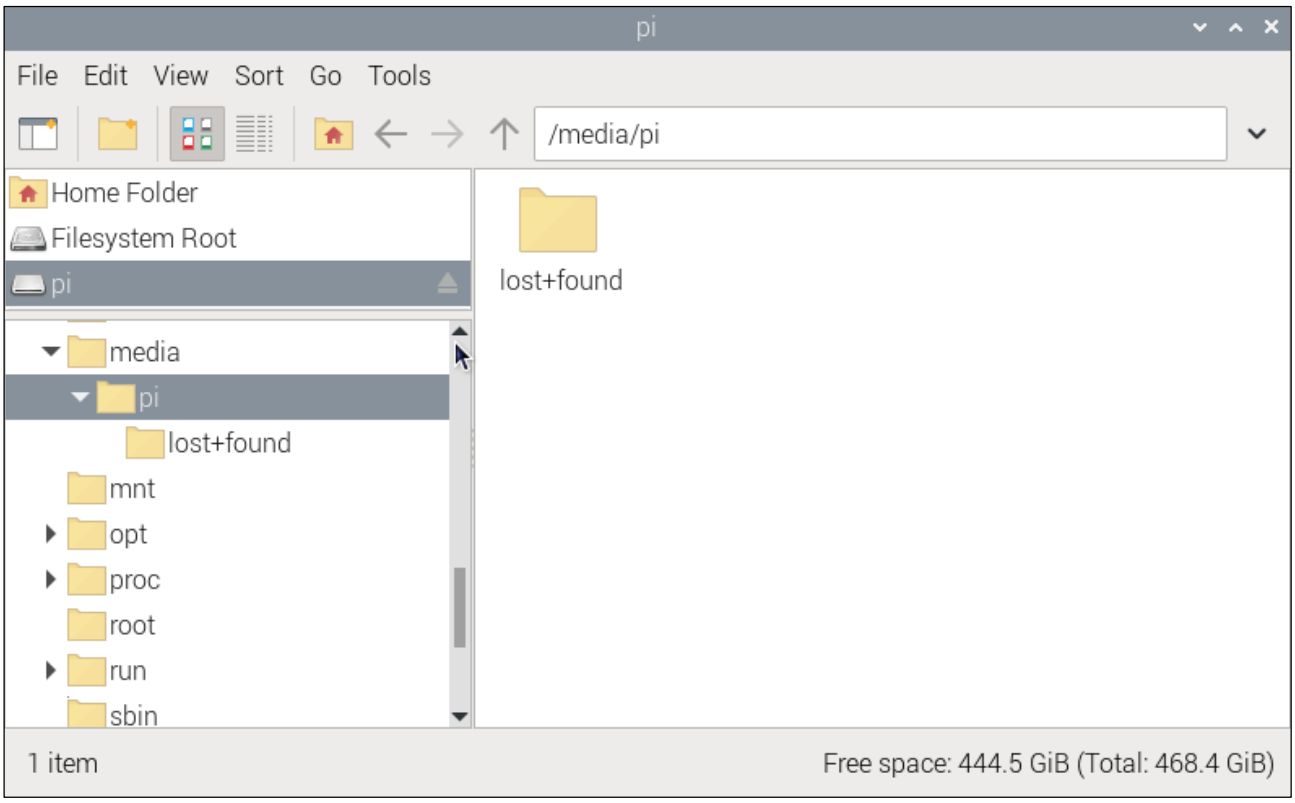


At this point, you can mount the disk using the mount command and then access the disk space through the file manager. Use the following command to mount the SSD:

```
mkdir pi
sudo mount /dev/nvme0n1p1 /media/pi
```

```
pi@raspberrypi:~ $ mkdir pi
pi@raspberrypi:~ $ sudo mount /dev/nvme0n1p1 /media/pi
```

Open the file manager, as shown below.



If you plan to use the SSD as a standard storage device, you can conclude the process here. However, if you want to further proceed with installing an operating system on the SSD, please read on.

2.4 Flashing the OS

Install the OS to SSD with the method similar to that in the previous section on installing a system onto an SD card. This time, operate on the Raspberry Pi.

Install rpi-imager with the following command:

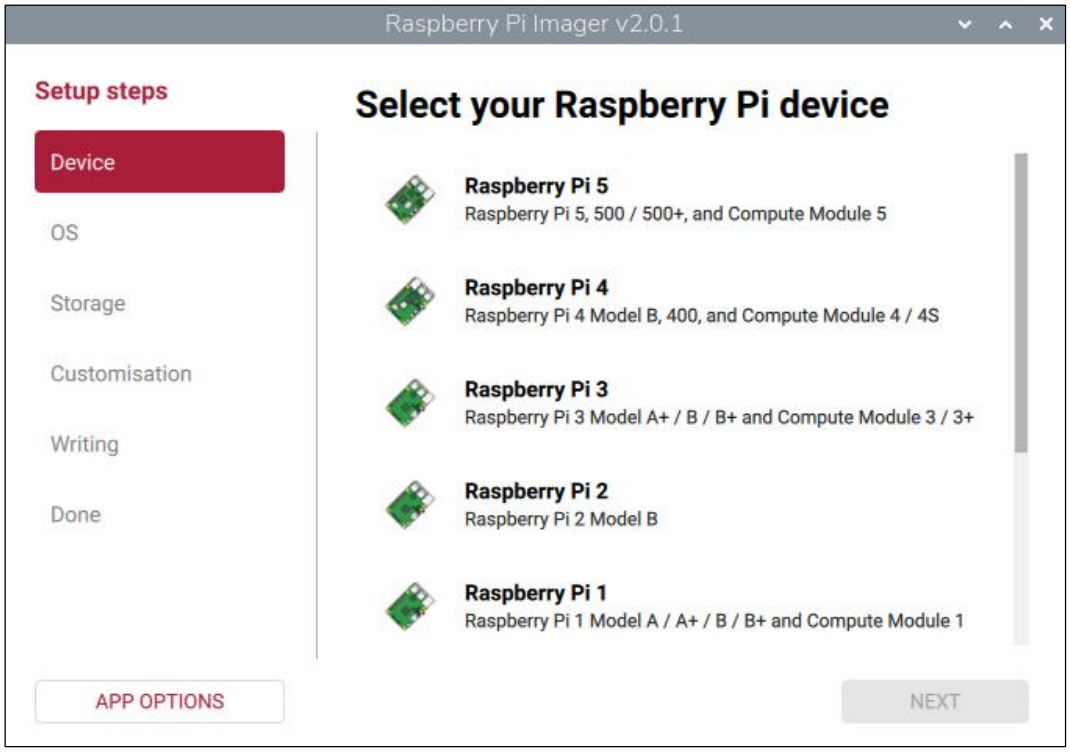
```
sudo apt install rpi-imager
```

```
pi@raspberrypi:~ $ sudo apt install rpi-imager
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
rpi-imager is already the newest version (1.8.5+rpt1).
0 upgraded, 0 newly installed, 0 to remove and 164 not upgraded.
pi@raspberrypi:~ $
```

Open rpi-imager:

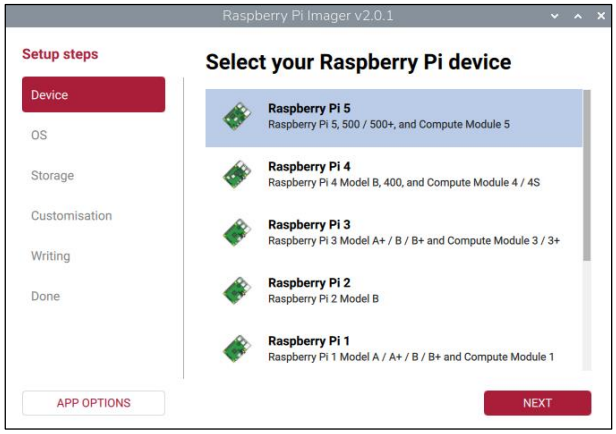
```
sudo rpi-imager
```

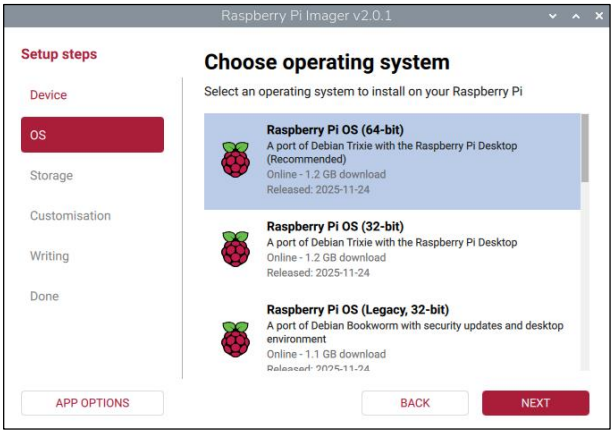
```
pi@raspberrypi:~ $ sudo rpi-imager
Running as root via sudo
Original user: pi
Original UID: 1000
Original home directory: /home/pi
DISPLAY already set to: :0
```

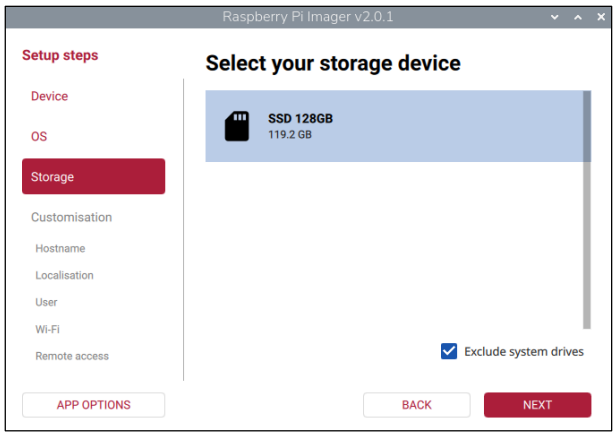


By this point, you should be quite familiar with the process.

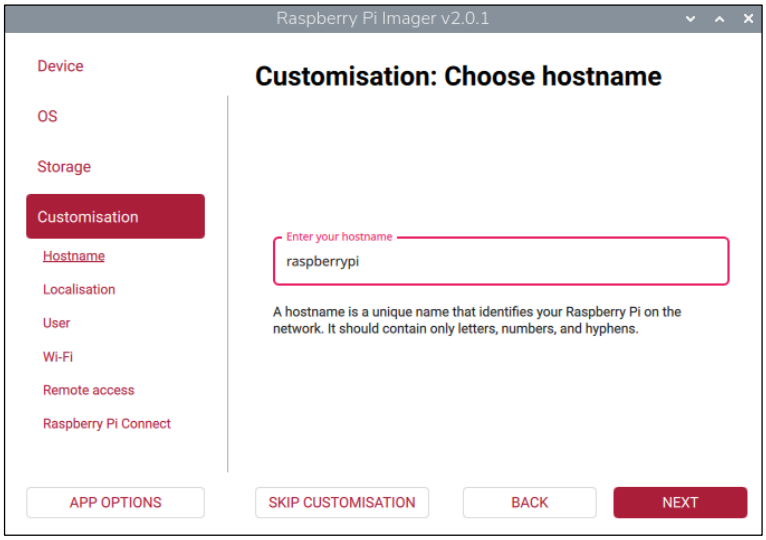
Select the Raspberry Pi 5 as your device and choose the online download for the operating system. (It is recommended to use a 64-bit Raspberry Pi system with recommended software). Choose your NVME SSD as the storage device. Click NEXT.



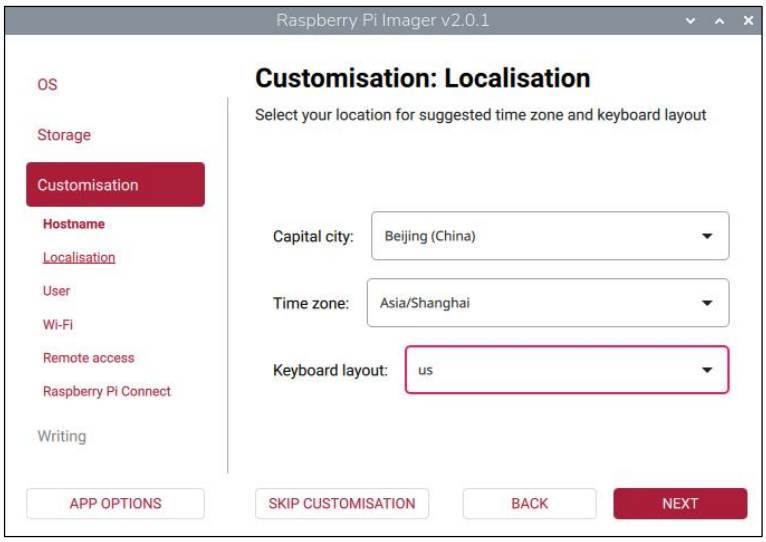




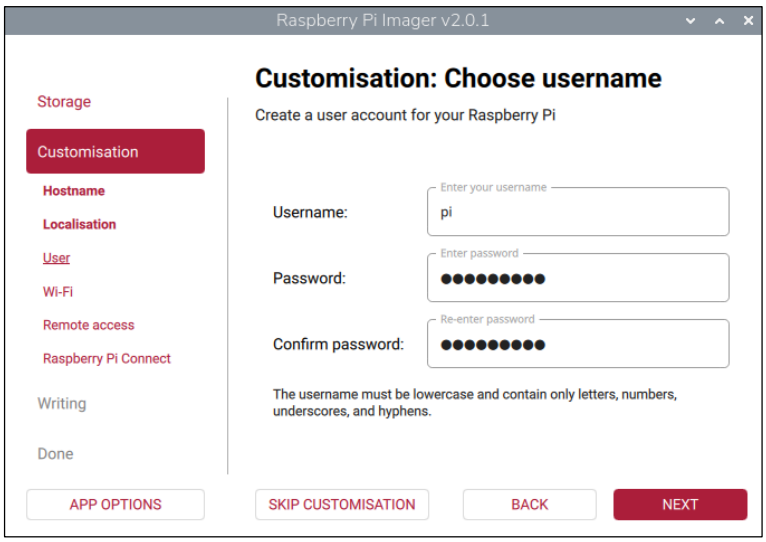
Enter the hostname.



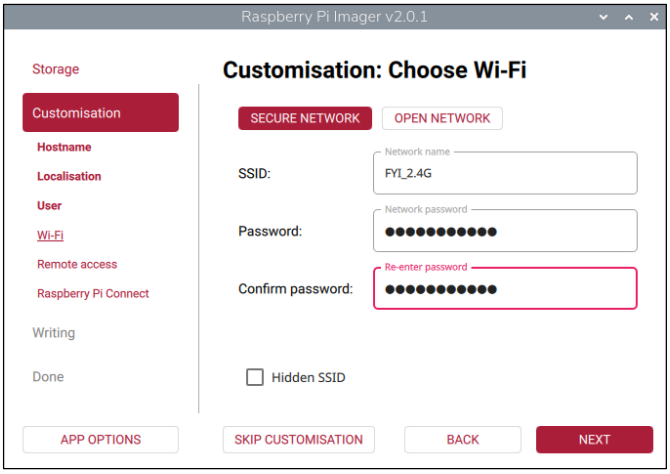
Select your location, timezone and keyboard layout. Then click NEXT.



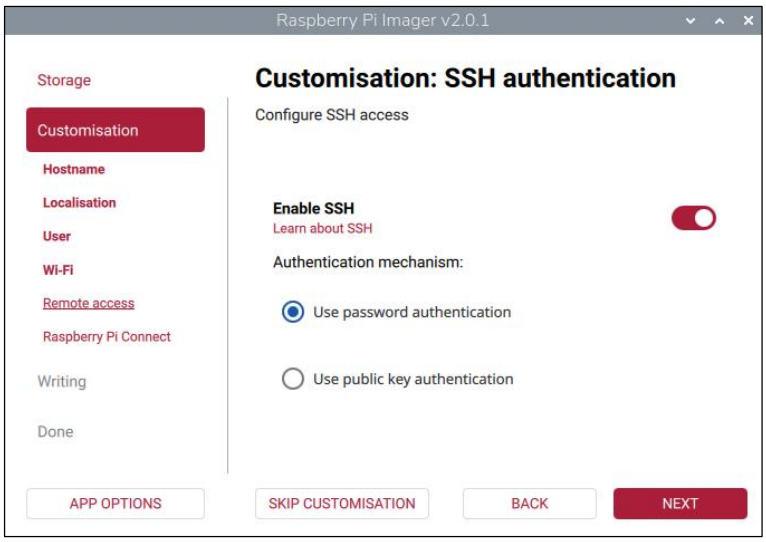
Configure your username and passowrd. The password needs to re-enter to confirm.



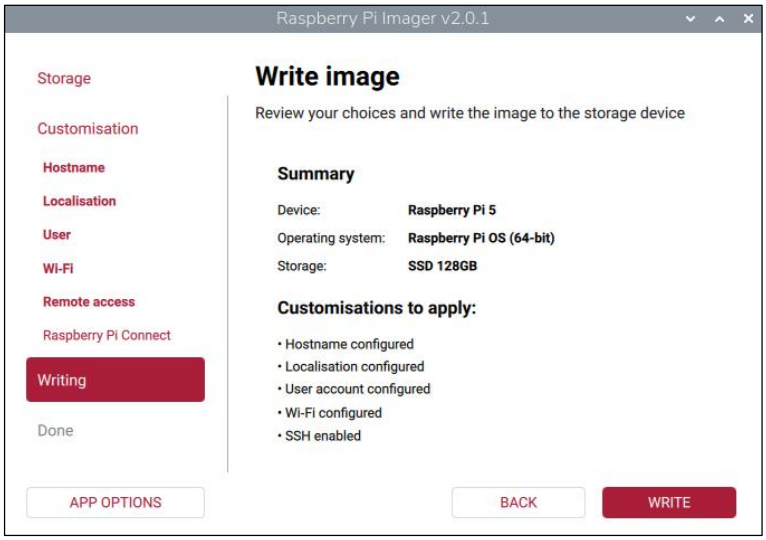
Step 4: Input the SSID and password of your wireless network. The passwords needs to re-enter to confirm. Then click NEXT.



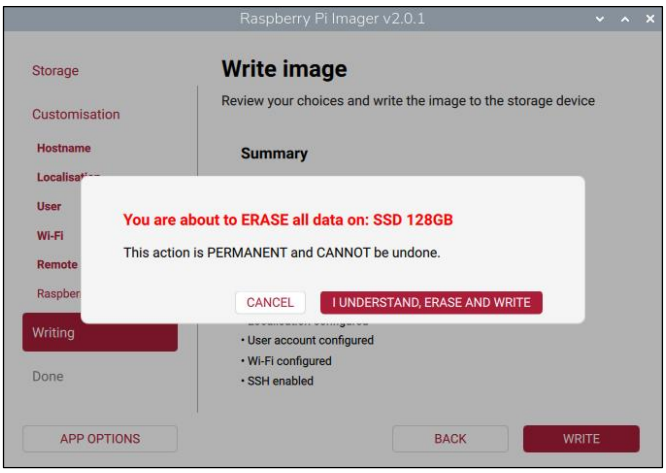
Enable SSH for ssh remote and click NEXT.



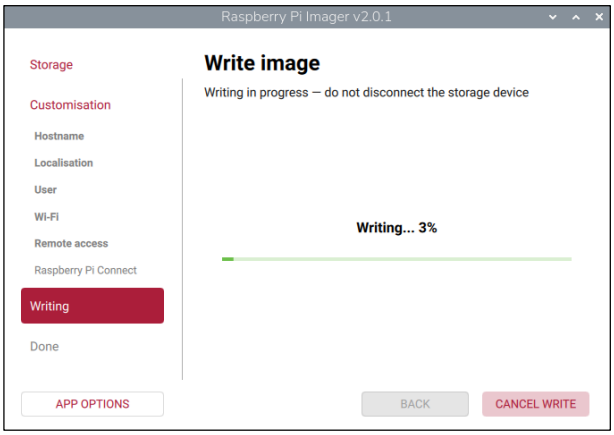
After finishing customisation configuration, the interface will show all the configuration you have made. Click **WRITE** to write the OS to your SD card after confirming everything is correct.

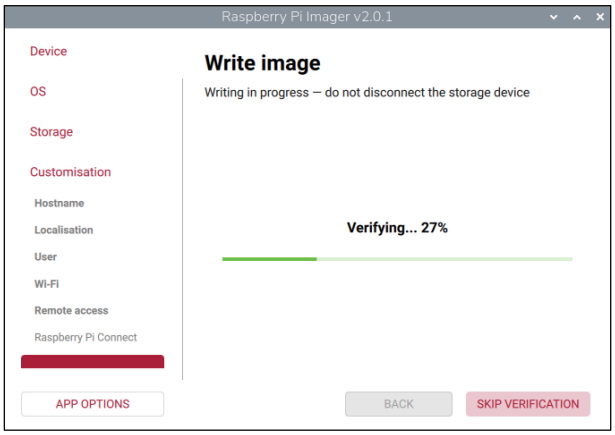


Warning: Before starting the write process, ensure you have backed up all data on the storage device. This operation will permanently erase all data on the device. If you have completed the backup, click **I UNDERSTAND, ERASE AND WRITE** to continue.

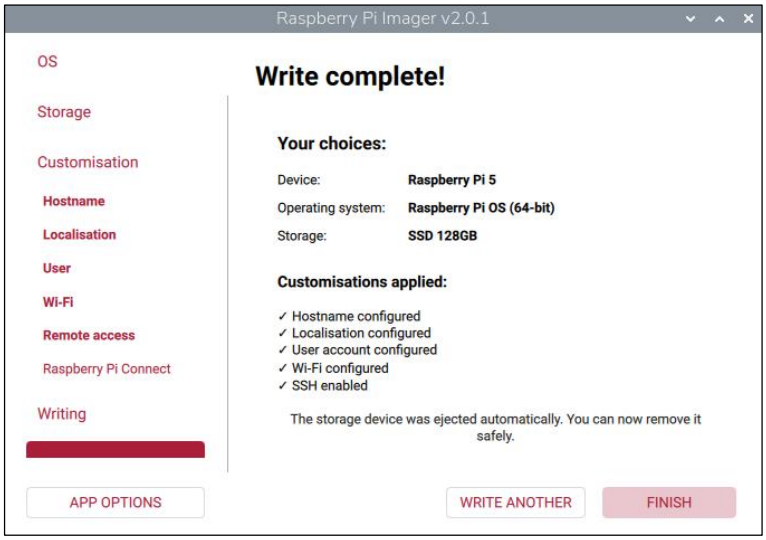


Wait for it to finish writing and verifying.





Congratulations! You have done the trickiest and the time-consuming part. Now that you have successfully installed the operating system onto the NVMe SSD, you are very close to achieving a triumph.



Next, boot into the system from SSD.

2.5 Enable PCIe 3.0 (on system written into SSD)

Note: This section does not apply to the Dual-NVMe Adapter Board or the Quad-NVMe Adapter Board, as they do not support the PCIe 3.0 protocol. If enabled, it may fail to boot from SSD.

If you have confirmed that SSD is with Phison controller in step 2, then you also need to enable PCIE3.0 on the system written into SSD.
If the controller of your SSD is not from Phison, **you can skip this section.**

The operation is as below:
Run the command lsblk to check the partitions of the SSD with Raspberry Pi OS written, as shown below:

```
pi@raspberrypi:~ $ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
mmcblk0     179:0    0  29.3G  0 disk
├mmcblk0p1 179:1    0   512M  0 part /boot/firmware
└mmcblk0p2 179:2    0  28.8G  0 part /
nvme0n1     259:0    0 119.2G  0 disk
├nvme0n1p1 259:1    0   512M  0 part
└nvme0n1p2 259:2    0 118.7G  0 part
pi@raspberrypi:~ $
```

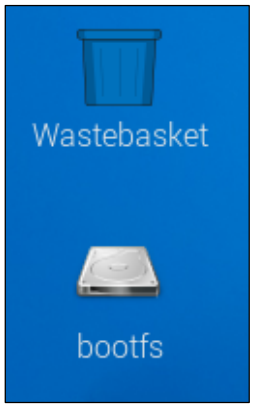
(The above screenshot is the result of a 128GB SSD with Phison as main controller.)

Run the following commands one by one to mount partition 1 of the SSD to the directory of /media/pi.

```
sudo mkdir /media/pi
sudo mount /dev/nvme0n1p1 /media/pi
```

```
pi@raspberrypi:~ $ sudo mkdir /media/pi
pi@raspberrypi:~ $ sudo mount /dev/nvme0n1p1 /media/pi
```

If it mounts successfully, you'll see the following disk icon on the desktop.



Open and modify the config.txt file with the following command.

```
sudo nano /media/pi/config.txt
```

```
pi@raspberrypi:~ $ sudo nano /media/pi/config.txt
```

Add the line dtparam=pciex1_gen=3 to the end of the file, as shown below:

```
# Disable compensation for displays with overscan
disable_overscan=1

# Run as fast as firmware / board allows
arm_boost=1

[cm4]
# Enable host mode on the 2711 built-in XHCI USB controller.
# This line should be removed if the legacy DWC2 controller is required
# (e.g. for USB device mode) or if USB support is not required.
otg_mode=1

[cm5]
dtoverlay=dwc2,dr_mode=host

[all]
dtparam=pciex1_gen=3
```

Press Ctrl-O to save the file, Enter to confirm, and Ctrl-X to exit.

2.6 Booting from SSD

After finishing flashing the OS to SSD, shutdown Raspberry Pi, **remove the power supply, and remove the SD card. Then connect the power, the Raspberry Pi will boot from SSD**.

The default boot order of Raspberry Pi is SD card → SSD → USB, Therefore, when the SD card is removed, the Raspberry Pi cannot detect the SD card, it will boot from SSD. By far, the Raspberry Pi can boot successfully from NVME SSD.

```
B1 SD Card Boot  Boot from SD Card before trying NVMe and then USB (RECOMMENDED)
B2 NVMe/USB Boot Boot from NVMe before trying USB and then SD Card
B3 Network Boot  Boot from Network unless override by SD Card
```

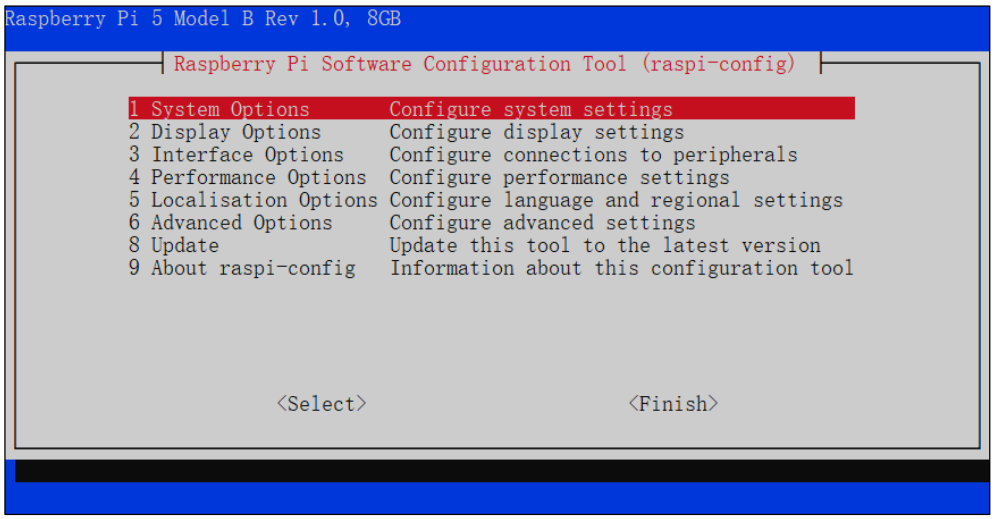
If you want the Raspberry Pi to boot from the SSD first, please continue with the following steps to modify the boot order. The boot order is saved in the Pi's EEPROM, so it does not matter whether you modify the boot order on SD card system or SSD system.

If you do not want to change the boot order, please skip this chapter.

Configuring the Boot Order

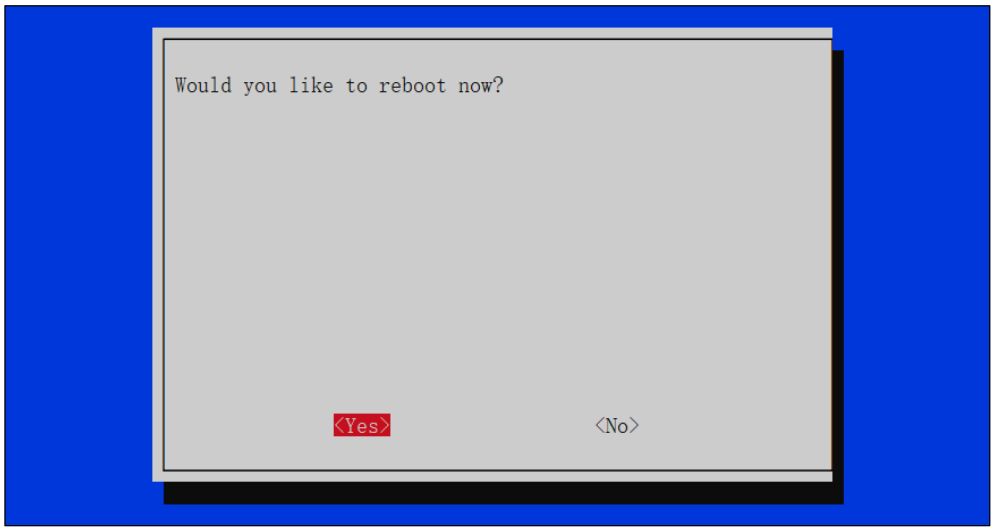
Type the following command in the Terminal.

```
sudo raspi-config
```



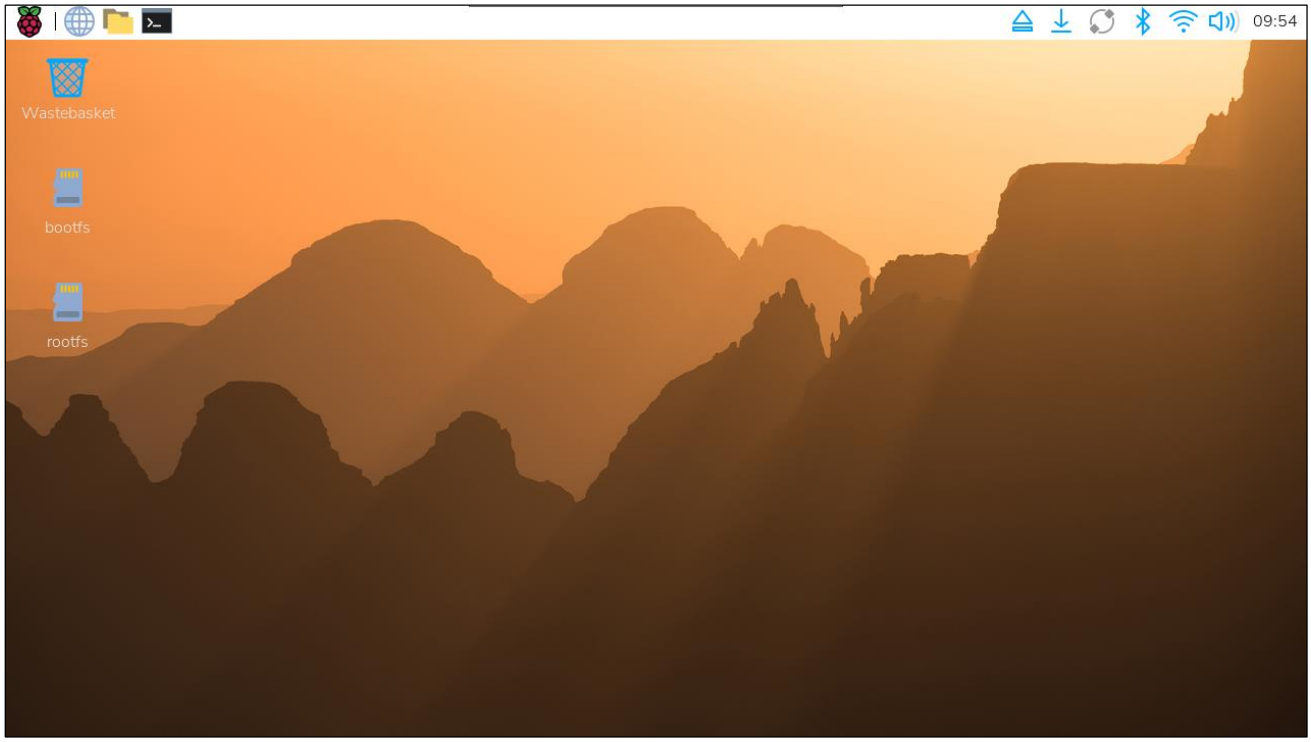
Using the keyboard's arrow keys and the Enter key, select the options in sequence.
"6 Advanced Options" →"A4 Boot Order"→"B2 NVME/USB Boot …"

Select "OK"→"Finish"→"Yes", and reboot your Raspberry Pi.



At this point, upon restarting, the Raspberry Pi will boot from the NVME SSD first. If you are using an external monitor, you will see that the Raspberry Pi has booted up correctly. If your SD card is still inserted, you will also see an icon on the desktop as shown below.

With this, the process of booting the Raspberry Pi from the NVME SSD has been fully completed.



If you use VNC viewer, you will need to repeat the previous steps to activate the VNC service as it is not yet enabled in the new system on the SSD. Here, we take Windows as an example.

Run the following command:

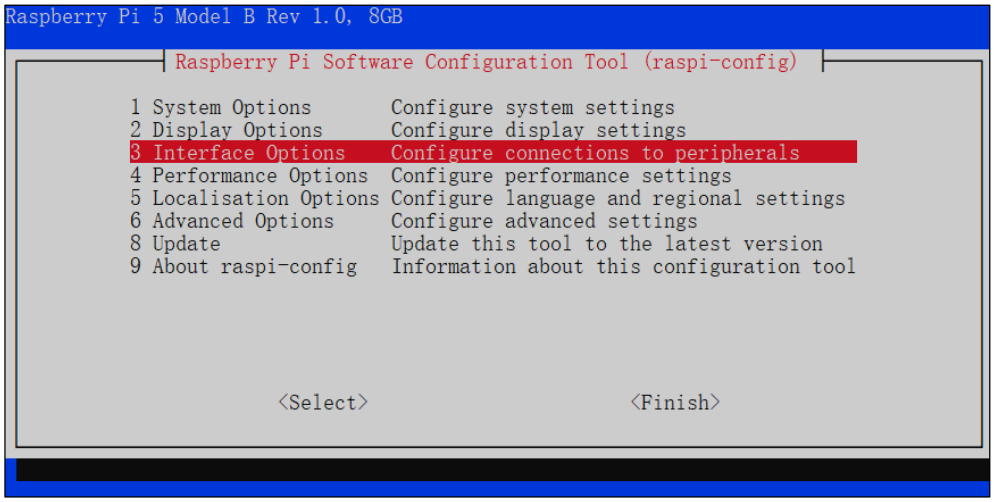
```
ssh pi@raspberrypi.local
```

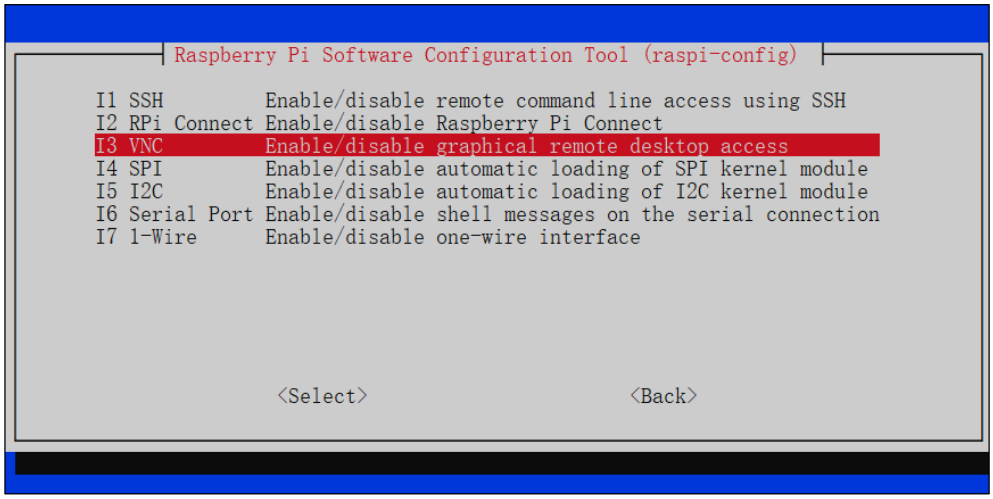
```
C:\Users\Administrator>ssh pi@raspberrypi.local
The authenticity of host 'raspberrypi.local (240e:3b4:3812:1fc0:954e:f55f:a772:fed5)' can't be established.
ECDSA key fingerprint is SHA256:hcx7u6H73nUsIc5WXA3HWa5GPSZEDroiz/mMbQx3ogc.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'raspberrypi.local,240e:3b4:3812:1fc0:954e:f55f:a772:fed5' (ECDSA) to the list of known hosts
.
pi@raspberrypi.local's password:
Linux raspberrypi 6.6.20+rpt-rpi-2712 #1 SMP PREEMPT Debian 1:6.6.20-1+rpt1 (2024-03-07) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Jun  3 16:50:25 2024

SSH is enabled and the default password for the 'pi' user has not been changed.
This is a security risk - please login as the 'pi' user and type 'passwd' to set a new password.

pi@raspberrypi:~ $
```

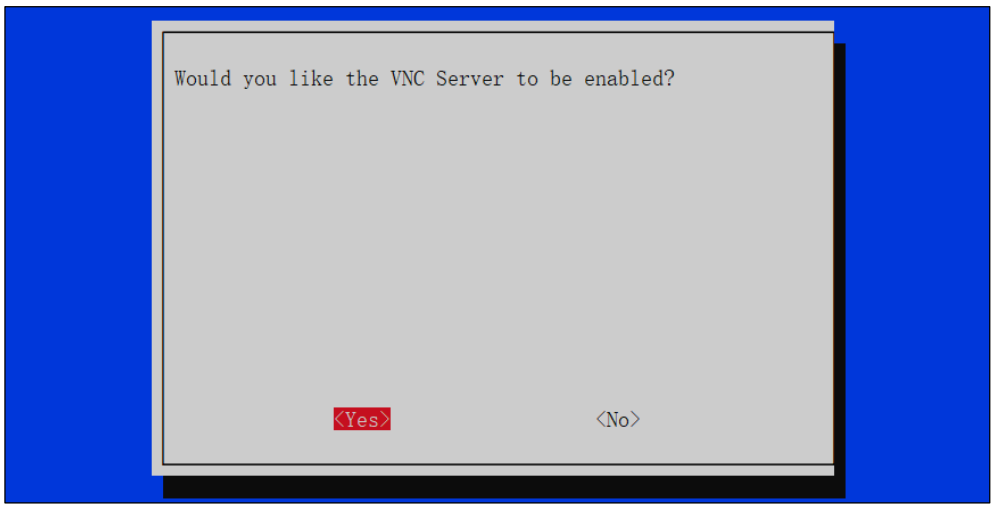
Once successfully ssh into Raspberry Pi, run the following command to open the configuration and enable VNC.

```
sudo raspi-config
```

Select “3 Interface Options”→”I2 VNC”→”Yes”→”Finish”.







Now you should be able to access Raspberry Pi via VNC.